

DIFFUSE GAUSSIAN TRUNCATION FOR DETERMINISTIC APPROXIMATE COUNTING

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ABSTRACT. We give deterministic fully polynomial-time approximation schemes for two dense counting problems on which the known deterministic algorithms, based on zero-free interpolation, run in quasipolynomial time. For fixed $0 < \gamma < 1/2$ and $0 < \theta \leq 1$, the first approximates $\text{haf}(A)$ for a symmetric matrix A of even order n when its support graph G has minimum degree at least $(1/2 + \gamma)n$ and its nonzero entries lie in $[\theta, 1]$. It also approximates permanents under the analogous bipartite condition, including all full-support matrices with entries in $[\theta, 1]$. For fixed $\beta > 0$ and $0 < \kappa \leq 1$, the second approximates the zero-field Ising partition function $Z(J)$ for zero-diagonal real symmetric matrices J satisfying $\max_{i,j} |J_{ij}| \leq \beta/n$ and $\lambda_{\max}(J) \leq 1 - \kappa$. This spectral condition is one-sided, with no separate lower-eigenvalue hypothesis. We further prove $\log \text{haf}(A) = h_A(G) - n/2 + O_{\gamma,\theta}(1)$ and $Z(J) = 2^n \det(I - J)^{-1/2} (1 + O_{\beta,\kappa}(1/n))$. Here $h_A(G)$ is the maximum weighted fractional-matching entropy. For unweighted graphs, the first formula improves the Cuckler–Kahn error from $o(n)$ to $O_\gamma(1)$ on the fixed-margin class. It also extends their formula to weights in $[\theta, 1]$.

Both algorithms follow from a common Gaussian truncation principle. Each problem reduces to integrals of products of a fixed entire function over Gaussian coordinates with a possibly indefinite moment matrix whose entries are $O(1/n)$. After linear cancellation and exact quadratic resummation, the coordinate remainder vanishes to order at least three. Complex dilation controls small supports. For large supports, we bound the recombined tail by a large-deviation estimate whose rate beats the entropy of the subsets. Together these give truncation error $(CR/n)^{R/2} + e^{-cn}$. This faster-than-geometric decay permits a cutoff satisfying $R \log(en/R) = O(\log n + \log(1/\varepsilon))$, which makes the enumeration polynomial. The retained terms are evaluated by monomer–dimer recurrences and one-dimensional quadrature for matchings, and by small spin sums for Ising.

1. INTRODUCTION

For the permanent and the Ising partition function on the dense inputs considered below, known deterministic approximation algorithms remain quasipolynomial. The Taylor expansion behind zero-free interpolation converges geometrically, so attaining accuracy ε requires $\Theta(\log(1/\varepsilon))$ layers, while evaluating through layer R costs $n^{\Theta(R)}$. We take a different route. After passing to a Gaussian product with a diffuse moment matrix, whose entries are $O(1/n)$, we exactly resum its quadratic part. The contribution supported on R coordinates is then suppressed by $(R/n)^{\alpha R}$ for some fixed $\alpha > 0$. This faster decay permits a cutoff with only polynomially many retained terms.

The resulting principle gives deterministic FPTASs for dense weighted perfect matchings and permanents, and for zero-field Ising models with diffuse couplings. The two applications arrive at the Gaussian product in different ways. For matchings, maximum-entropy scaling removes the main exponential contribution, and an inverse-Gamma identity produces a one-parameter family of Gaussian products. For Ising, the Hubbard–Stratonovich identity gives a Gaussian product directly. In both cases the same completion exposes the higher-order remainder to which the truncation theorem applies.

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Main results. We first state the matching result. Let K_n be the complete graph on n vertices. If A is a symmetric matrix of even order n , its hafnian is

$$\text{haf}(A) = \sum_M \prod_{uv \in M} A_{uv},$$

where the sum is over the perfect matchings M of K_n . The support graph of a nonnegative symmetric matrix A has an edge uv exactly when $A_{uv} > 0$. For a graph G , write $\delta(G)$ for its minimum degree and $\# \text{PM}(G)$ for its number of perfect matchings. Hence $\# \text{PM}(G)$ is the hafnian of the adjacency matrix of G .

For the bipartite problem, let S_m be the permutations of $\{1, \dots, m\}$. The permanent of an m by m matrix $B = (B_{ij})$ is

$$\text{per}(B) = \sum_{\sigma \in S_m} \prod_{i=1}^m B_{i, \sigma(i)}.$$

The bipartite support of a nonnegative matrix has an edge ij exactly when $B_{ij} > 0$. For a zero-one matrix, the permanent counts the perfect matchings of this support graph.

We measure multiplicative error by $e^{\pm \varepsilon}$. Thus an estimate $\hat{Q}$ of a positive quantity Q is acceptable when $e^{-\varepsilon}Q \leq \hat{Q} \leq e^{\varepsilon}Q$. This convention is equivalent to $1 \pm O(\varepsilon)$ for $0 < \varepsilon < 1$. All numerical inputs, including ε , are encoded in binary. For a rational matrix C , let L_C denote its encoding length, and let L_ε denote the encoding length of ε .

Theorem 1.1 (Main result for matchings). *Fix $0 < \gamma < 1/2$ and $0 < \theta \leq 1$. The following deterministic algorithms exist.*

- (a) *Let A be a rational symmetric n by n matrix, where n is even, with zero diagonal and all off-diagonal entries in $\{0\} \cup [\theta, 1]$. Suppose that its support graph G satisfies*

$$(1) \quad \delta(G) \geq \left(\frac{1}{2} + \gamma\right)n,$$

and let $0 < \varepsilon < 1$. The algorithm returns an estimate $\hat{H} > 0$ such that

$$e^{-\varepsilon} \text{haf}(A) \leq \hat{H} \leq e^{\varepsilon} \text{haf}(A).$$

Its bit complexity is $n^{O_{\gamma, \theta}(1)} \varepsilon^{-O_{\gamma, \theta}(1)} \text{poly}(L_A + L_\varepsilon)$.

- (b) *Let B be a rational m by m matrix with entries in $\{0\} \cup [\theta, 1]$. Suppose that every row and column of its support contains at least $(1/2 + \gamma)m$ entries, and let $0 < \varepsilon < 1$. The algorithm returns an estimate $\hat{Z}_B > 0$ such that*

$$e^{-\varepsilon} \text{per}(B) \leq \hat{Z}_B \leq e^{\varepsilon} \text{per}(B).$$

Its bit complexity is $m^{O_{\gamma, \theta}(1)} \varepsilon^{-O_{\gamma, \theta}(1)} \text{poly}(L_B + L_\varepsilon)$.

Taking $\theta = 1$ gives a deterministic FPTAS for unweighted perfect matchings in the corresponding graph classes. Dirac's theorem guarantees a perfect matching at minimum degree $n/2$ [Dir52]. The fixed margin in Theorem 1.1 is therefore measured from the natural existence threshold, although it still allows each vertex to miss nearly half of the other vertices.

We next state the Ising result. Let J be a real symmetric n by n matrix with zero diagonal. For a spin configuration $\sigma \in \{\pm 1\}^n$, its interaction energy is $\frac{1}{2} \sigma^\top J \sigma$, and the zero-field partition function is $Z(J) = \sum_{\sigma \in \{\pm 1\}^n} \exp(\sigma^\top J \sigma / 2)$. We write $\lambda_{\max}(J)$ for the largest eigenvalue of J , use I for the identity matrix, and let $\|\cdot\|$ denote the spectral operator norm.

Theorem 1.2 (Diffuse zero-field Ising model). *Fix $\beta > 0$ and $0 < \kappa \leq 1$. Let J be a real symmetric n by n matrix with rational entries and zero diagonal, and suppose that*

$$\max_{i,j} |J_{ij}| \leq \frac{\beta}{n}, \quad \lambda_{\max}(J) \leq 1 - \kappa.$$

Given $0 < \varepsilon < 1$, there is a deterministic algorithm that returns a positive estimate $\widehat{Z}$ satisfying

$$e^{-\varepsilon} Z(J) \leq \widehat{Z} \leq e^{\varepsilon} Z(J).$$

Its bit complexity is $n^{O_{\beta,\kappa}(1)} \varepsilon^{-O_{\beta,\kappa}(1)} \text{poly}(L_J + L_\varepsilon)$. Moreover, uniformly over all real matrices satisfying the two displayed hypotheses,

$$Z(J) = 2^n \det(I - J)^{-1/2} (1 + O_{\beta,\kappa}(1/n)),$$

where the square root is positive. In particular, $\log Z(J) = n \log 2 - \frac{1}{2} \log \det(I - J) + O_{\beta,\kappa}(1/n)$. The determinant term satisfies $|\frac{1}{2} \log \det(I - J)| \leq \beta^2/(4\kappa)$.

Rationality in [Theorem 1.2](#) is required only for the algorithmic statement.

The spectral assumption in [Theorem 1.2](#) is one-sided. It keeps the top eigenvalue a fixed distance below the Gaussian singularity at 1, but it does not require $\|J\| < 1$. Negative eigenvalues may therefore have magnitude larger than one. The entrywise hypothesis is separate. It is the diffuseness condition that makes the coordinate-subset expansion summable.

We record three consequences of the matching theorem. The first removes support zeros altogether.

Corollary 1.3 (Full-support inputs with weights bounded away from zero). *Fix $0 < \theta \leq 1$. There is a deterministic FPTAS for the hafnian of every even-dimensional rational symmetric matrix whose diagonal is zero and whose off-diagonal entries lie in $[\theta, 1]$, and for the permanent of every rational square matrix whose entries lie in $[\theta, 1]$.*

The same algorithm treats odd-order graphs by leaving one vertex unmatched.

Corollary 1.4 (Near-perfect matchings). *Fix $0 < \gamma < 1/2$. There is a deterministic FPTAS for the number of near-perfect matchings in every odd-order graph G satisfying $\delta(G) \geq (1/2 + \gamma)|V(G)|$.*

The interval $[\theta, 1]$ is only a normalization. If all positive entries lie in $[c_0, C_0]$ for fixed positive endpoints, divide the matrix by C_0 and restore the resulting homogeneous factor at the end.

The third consequence identifies the leading exponential contribution to a weighted hafnian. Suppose that $A = (a_{uv})$ satisfies part (a) of [Theorem 1.1](#), and let G be its support graph. Define the fractional perfect-matching polytope

$$\mathcal{D}(G) = \left\{ w \in \mathbb{R}_{\geq 0}^{E(G)} : \sum_{e \ni v} w_e = 1 \text{ for every } v \right\}.$$

Thus a point of $\mathcal{D}(G)$ assigns nonnegative weights to the edges, with total weight one at each vertex. Its weighted entropy is

$$h_A(G) = \max_{w \in \mathcal{D}(G)} \sum_{uv \in E(G)} w_{uv} \log \frac{a_{uv}}{w_{uv}}.$$

All logarithms are natural, and a zero summand is interpreted as zero. For the adjacency matrix of G , this is the entropy $h(G)$ used by Cuckler and Kahn.

Corollary 1.5 (Entropy formula). *Fix $0 < \gamma < 1/2$ and $0 < \theta \leq 1$. There are constants $0 < a_{\gamma,\theta} \leq b_{\gamma,\theta} < \infty$ such that every matrix A satisfying part (a) of [Theorem 1.1](#) satisfies*

$$a_{\gamma,\theta} \leq \frac{\text{haf}(A)}{e^{h_A(G)} (n-1)!! (n-1)^{-n/2}} \leq b_{\gamma,\theta}.$$

Here $(n-1)!! = 1 \cdot 3 \cdots (n-1)$. Equivalently, $\log \text{haf}(A) = h_A(G) - n/2 + O_{\gamma,\theta}(1)$, where the remainder is bounded independently of n and A .

For unweighted Dirac graphs, Cuckler and Kahn determine the logarithm of the number of perfect matchings up to an $o(n)$ error [[CK09a](#), [CK09b](#)]. In the fixed-margin regime, [Corollary 1.5](#) improves this to $O_\gamma(1)$ and extends the formula to weights in a fixed positive interval. The algorithms allow their constants and running-time exponents to depend on the fixed parameters. We do not optimize this dependence.

Technical overview. The following overview traces both counting problems to one common truncation theorem, then explains the two estimates in its proof. The argument is organized into six stages, with the fourth divided into its two estimates.

1. *Entropy and the inverse-Gamma reduction.* Let A be an input to part (a) of [Theorem 1.1](#). Maximizing the weighted entropy gives a unique fractional perfect matching w^* . The Lagrange-multiplier equations factor it as $w_{uv}^* = a_{uv}r_ur_v$ for positive vertex factors r_v . Every perfect matching uses each factor once, so

$$\text{haf}(A) = e^{h_A(G)} \text{haf}(X), \quad X_{uv} = a_{uv}r_ur_v.$$

The matrix X is symmetric and stochastic. Let $\mathbf{J}_n$ and I_n be the all-ones and identity matrices, and let $\mathbf{1}$ be the all-ones column vector. Put $P_0 = (\mathbf{J}_n - I_n)/(n-1)$, and write $E = X - P_0$. The degree and weight margins make E diffuse and supply a fixed spectral margin. Since X and P_0 are stochastic, $E\mathbf{1} = 0$. Thus only the normalized ratio $Q_E = \text{haf}(P_0 + E)/\text{haf}(P_0)$ remains to be computed.

Expand the numerator according to the edges chosen from E . The coefficient left by the reference matrix depends only on the number of chosen edges and is a negative moment of one Gamma variable. More precisely, if U has shape $(n+1)/2$ and rate $(n-1)/2$, then

$$Q_E = \mathbb{E} Z_E(U^{-1}), \quad Z_E(t) = \sum_M t^{|M|} \prod_{uv \in M} E_{uv},$$

where M ranges over all matchings. The random variable U is concentrated near its mode 1, and the contribution outside a fixed interval around 1 is exponentially small. The bipartite argument begins with Sinkhorn scaling and applies the same construction after symmetric dilation, using the complete-bipartite reference and slightly different Gamma parameters.

2. *The Hubbard–Stratonovich reduction.* The starting point for [Theorem 1.2](#) is the Hubbard–Stratonovich identity, which removes the spin sum. If ψ is the canonical complex Gaussian vector with bilinear moment matrix J , and $\mathbb{E}_J$ denotes expectation with respect to this vector, summing the resulting exponential moments over the spins gives

$$Z(J) = 2^n \mathbb{E}_J \prod_{i=1}^n \cosh(\psi_i).$$

Unlike the matching problem, this reduction needs neither an entropy scaling nor an auxiliary Gamma average. It does not yet have the higher-order remainder required by the truncation theorem.

3. *Quadratic resummation.* The matching polynomial also has a Gaussian product representation. If ξ has bilinear second-moment matrix E , then the Wick–Isserlis formula [[Iss18](#)] gives $Z_E(t) = \mathbb{E} \prod_i (1 + \sqrt{t} \xi_i)$. A direct expansion cannot be truncated. A term supported on s coordinates has the rough size $(Cs/n)^{s/2}$, while there are about $(en/s)^s$ possible supports. The centering relation $E\mathbf{1} = 0$ creates cancellation between the raw layers, but taking their absolute values destroys it. To repair this, we remove the first two Taylor terms exactly. The centering relation kills the linear term, and the quadratic term is absorbed into the Gaussian density. The same completion removes the quadratic term of $\cosh z$ in the Ising product. For a real symmetric moment matrix L , let ϕ denote the canonical complex Gaussian vector with bilinear moments L . We write $\mathcal{Z}_L = \mathbb{E}_L \prod_i g(\phi_i)$, using the relevant one-coordinate factor in each application. The two exact identities are

$$\begin{aligned} Z_E(t) &= \det(I + tE)^{-1/2} \mathcal{Z}_{K_t}, & K_t &= tE(I + tE)^{-1}, \\ Z(J) &= 2^n \det(I - J)^{-1/2} \mathcal{Z}_K, & K &= J(I - J)^{-1}. \end{aligned}$$

The corresponding one-coordinate factors are

$$g_{\text{pm}}(z) = (1+z)e^{-z+z^2/2}, \quad g_{\text{Is}}(z) = \cosh(z)e^{-z^2/2}.$$

Now $g_{\text{pm}}(z) - 1 = O(z^3)$. In the Ising identity the same quadratic resummation leaves $g_{\text{Is}}(z) - 1 = -z^4/12 + O(z^6)$. Completion also preserves the entrywise scale and the required spectral margin. Consequently, both applications enter the common theorem, with vanishing order three for matchings and four for Ising.

For either factor, put $f = g - 1$ and write $[n] = \{1, \dots, n\}$. If $S \subseteq [n]$, let

$$W_K(S) = \mathbb{E}_{K[S]} \prod_{i \in S} f(\phi_i), \quad W_K(\emptyset) = 1,$$

where $K[S]$ is the principal submatrix indexed by S .

Theorem (Theorem 2.3, informal). *Suppose that g has a quadratic Gaussian envelope, that $f(z) = O(z^\ell)$ for some $\ell \geq 3$, and that, for fixed $\beta, d > 0$, the real symmetric moment matrix K satisfies $\max_{i,j} |K_{ij}| \leq \beta/n$ and the spectral margin condition of Section 2 with constant d . Put $\alpha = \ell/2 - 1$. Then there are constants $c, C, a > 0$ such that*

$$\mathcal{Z}_K = \sum_{S \subseteq [n]} W_K(S), \quad \left| \mathcal{Z}_K - \sum_{|S| < R} W_K(S) \right| \leq \left(C \left(\frac{R}{n} \right)^\alpha \right)^R + e^{-an}$$

for every $2 \leq R \leq cn$ and all sufficiently large n . Moreover, $\mathcal{Z}_K = 1 + O(1/n)$ uniformly over the same class.

The formal statement makes the envelope and spectral margin precise. It does not require K to be positive semidefinite, the coefficients to have one sign, or an associated polynomial to be zero-free. Once the input hypotheses produce an admissible pair (g, K) , the tail analysis is common to both applications.

4(a). *Small coordinate sets.* Let S have size s below a small fixed multiple of n . To bound $W_K(S)$, consider the complex dilation $\Phi_S(\zeta) = \mathbb{E}_{K[S]} \prod_{i \in S} f(\zeta \phi_i)$. The entrywise bound on K makes the row norm of $K[S]$ at most $\beta s/n$. Hence Φ_S is holomorphic and uniformly bounded on a disk of radius comparable with $\sqrt{n/s}$. Because f vanishes to order ℓ , the function Φ_S vanishes to order ℓs at the origin. The maximum-modulus principle then gives

$$|W_K(S)| \leq \left(\frac{Cs}{n} \right)^{\ell s/2}.$$

Summing over all S of size s gives the layer bound $(C(s/n)^\alpha)^s$, where $\alpha = \ell/2 - 1 > 0$. This proves the first term in the truncation estimate as long as the support remains below a small linear cutoff.

4(b). *Recombining the large tail.* At a linear cutoff the dilation radius is only a constant, so the preceding termwise bound no longer pays for all supports. This is the main difficulty. We return to the single Gaussian integral and use

$$\sum_{|S| \geq R} \prod_{i \in S} f(\phi_i) = \prod_{i=1}^n g(\phi_i) - \sum_{|S| < R} \prod_{i \in S} f(\phi_i).$$

The absolute value is taken only after this recombination. We then separate the event on which few coordinates of ϕ have magnitude greater than a small fixed constant $\delta > 0$ from the event on which many do.

On the first event, every large support contains many small coordinates, and the local estimate $|f(z)| \leq C_f \delta^{\ell-2} |z|^2$ gives an exponential contraction. An elementary-symmetric-function bound makes this contraction explicit, and δ is chosen so that it dominates the remaining combinatorial factor. The quadratic envelope is integrable by the spectral margin.

On the second event, some set T of r coordinates carries a fixed amount of Gaussian energy. The quadratic form measuring this energy is controlled by the principal submatrix $|K|[[T]]$, where $|K|$ is the spectral absolute value of K . Diffuseness yields the estimate

$$||K|[[T]]|| \leq \beta \sqrt{r/n},$$

which permits an exponential tilt of order $\sqrt{n/r}$. For some fixed $a > 0$, a fixed witness set then costs $\exp\{-a\sqrt{rn}\}$. Write $\rho = r/n$. When ρ is a small positive constant, this gain has rate $\sqrt{\rho}$, whereas the number of witness sets has entropy rate $\rho \log(1/\rho)$. Since $\rho \log(1/\rho) = o(\sqrt{\rho})$, choosing the linear cutoff small enough makes the total contribution exponentially small. This is why the present proof uses more than a global spectral bound, and why the entrywise hypothesis is essential to it.

5. Evaluating retained subsets. It remains to compute $W_K(S)$ for $|S| < R$. Since $f = g - 1$, inclusion–exclusion expresses it through full-product expectations on subsets $T \subseteq S$. For matchings, each such expectation is a weighted monomer–dimer partition function and is evaluated by vertex deletion. For Ising, it is an explicit sum over the $2^{|T|}$ spin configurations on T . In both cases, one coefficient $W_K(S)$ costs $3^{|S|}$ poly($|S|$) arithmetic operations.

6. From truncation to an FPTAS. Let b be of order $\log_2(1/\varepsilon)$, and fix a sufficiently large constant C_{cut} . We choose the least integer $R \geq 2$ for which

$$\left(C_{\text{cut}} \left(\frac{R}{n}\right)^\alpha\right)^R \leq 2^{-b}.$$

Minimality gives $R \log(en/R) = O(b + \log n)$, and therefore the sum of the retained costs is $n^{O(1)} 2^{O(b)}$. If one simply took R proportional to b , coefficient enumeration would instead have the quasipolynomial form $n^{O(\log(1/\varepsilon))}$. The least-cutoff balance is what turns the truncation estimate into a fully polynomial scheme. When b is a fixed positive multiple of n , direct enumeration costs $2^{O(n)} = 2^{O(b)}$ and supplies the complementary branch.

The zero-free interpolation algorithms cited below use logarithmically many Taylor coefficients, but their dense-instance running times are quasipolynomial [Bar17, BB21]. Our completion resums the determinant contribution exactly, and the remaining $(R/n)^{\alpha R}$ decay makes the adaptive enumeration polynomial.

Every retained term is an explicit formula involving matrices that remain a fixed distance from singularity, so polynomially many working bits suffice. The details, including the entropy scaling, are given in [Appendix D](#). The Ising application needs no scaling step.

Previous work. *Randomized matching algorithms.* Exact counting is #P-complete already for zero-one permanents [Val79], and remains #P-complete on very dense bipartite and general graphs [MW22]. Jerrum and Sinclair gave an FPRAS for graphs of even order n and minimum degree at least $n/2$ [JS89], and Jerrum, Sinclair, and Vigoda gave an FPRAS for the permanent of every nonnegative matrix [JSV04]. Chen, Vigoda, and Yang recently improved the running time in this line of work [CVY26]. For general nonbipartite graphs, an FPRAS for the number of perfect matchings remains open. Ebrahimnejad, Nagda, and Oveis Gharan obtained randomized approximate counting and sampling for regular strong expanders [ENOG22].

Deterministic matching algorithms. For matrices with entries in $[\theta, 1]$, Barvinok gives a deterministic quasipolynomial-time relative approximation for permanents and hafnians [Bar17]. The same work treats zero-one permanents with a small fixed fraction of zeros in each row and column. The full-support case of [Theorem 1.1](#) makes the running time fully polynomial, and its support version reaches every fixed positive margin above the Dirac threshold. For arbitrary nonnegative permanents, deterministic methods give much coarser universal factors. One example is the tight $(\sqrt{2})^m$ guarantee for the Bethe permanent [AR25]. These results apply much more broadly, but they do not give arbitrary relative accuracy.

Dense matching asymptotics. Cuckler and Kahn proved $\log \# \text{PM}(G) = h(G) - n/2 + o(n)$ for every even-order Dirac graph [CK09a, CK09b]. Corollary 1.5 replaces $o(n)$ by a bounded remainder on the fixed-margin class. McCullagh obtained a determinantal approximation with relative error $O(1/m)$ for moderate-deviation sequences of doubly stochastic matrices [McC14]. Koehler and Leung prove high-probability zero-free regions for permanents of an all-ones matrix under independent centered perturbations [KL26]. Their theorem is an average-case bipartite result, while the hypotheses here are deterministic and also cover nonbipartite hafnians.

Ising algorithms. Jerrum and Sinclair gave an FPRAS for ferromagnetic Ising systems [JS93]. Eldan, Koehler, and Zeitouni proved rapid Glauber mixing for general interactions under the spectral condition $\|J\| < 1$ [EKZ22]. On the diffuse part of that norm regime, Theorem 1.2 gives deterministic relative approximation, and its one-sided condition also permits negative eigenvalues of magnitude larger than one. On the deterministic side, complex-zero methods give an FPTAS for bounded-degree zero-field Ising models throughout the correlation-decay region [LSS19]. More generally, Patel and Regts turn zero-free regions into deterministic polynomial-time approximation algorithms for a broad class of bounded-degree graph polynomials [PR17]. Their coefficient computation enumerates bounded-size connected subgraphs and relies on the maximum degree being bounded. Barvinok and Barvinok give a quasipolynomial relative approximation under a global bound on the total absolute interaction incident to each spin [BB21]. On dense instances, its Taylor truncation has geometric decay and quasipolynomial coefficient-enumeration cost. Mean-field methods apply much more broadly to dense models, but give additive control of the free energy rather than relative control of the partition function [JKM18].

The paper is organized as follows. Section 2 states and proves the common truncation theorem. Section 3 develops the matching and permanent applications. The Ising application is proved in Section 4. We return to the limitations of the method and the problems they leave open in Section 5. The appendices contain the deferred scaling, analytic, and bit-complexity estimates.

2. A TRUNCATION PRINCIPLE FOR DIFFUSE GAUSSIAN PRODUCTS

2.1. Setup and theorem. Although the matching and Ising reductions begin differently, after quadratic completion they produce the same analytic object: the expectation of a product $\prod_i g(\phi_i)$, where g is entire and the Gaussian bilinear moment matrix has entries of order $1/n$. The moment matrix need not be positive semidefinite, so the coordinates of ϕ may be complex. Wick's rule still expresses their moments in terms of this matrix, but absolute convergence must be controlled separately. The quadratic envelope and spectral margin introduced below provide that control uniformly over the admissible class. The theorem of this section shows that if $g - 1$ vanishes to order at least three, then the exact expansion by coordinate support has a rapidly decaying tail. In the two applications, the matching factor grows in the real direction and the Ising factor in the imaginary one.

For a real symmetric matrix K , write $K = K_+ - K_-$ for its positive and negative parts, and put

$$|K| = K_+ + K_-, \quad A_K = K_+^{1/2} + iK_-^{1/2}.$$

The matrix A_K is complex symmetric, $A_K^2 = K$, and $A_K^* A_K = |K|$. If x is a standard real Gaussian vector, we write $\phi = A_K x$. Thus $\mathbb{E} \phi \phi^\top = K$ and

$$\sum_i (\text{Re } \phi_i)^2 = x^\top K_+ x, \quad \sum_i (\text{Im } \phi_i)^2 = x^\top K_- x.$$

Whenever the integral converges absolutely, let $\mathbb{E}_K F(\phi) = \mathbb{E}_x F(A_K x)$. For a positive integer n , write $[n] = \{1, \dots, n\}$. For $S \subseteq [n]$, let $K[S]$ denote the principal submatrix indexed by S . For a matrix C , write $\|C\|_{\text{row}} = \max_i \sum_j |C_{ij}|$. We use

$$h_2(u) = -u \log u - (1 - u) \log(1 - u)$$

for the binary entropy, with the usual continuous interpretation at zero and one.

Together with the spectral margin imposed below, the global bound in the next definition makes the product integrable, with enough room left for the exponential tilts used below. The local vanishing order counts how many Gaussian legs each selected coordinate must carry.

Definition 2.1 (Admissible scalar factor). Let $a_R, a_I \geq 0$ be two envelope weights, not both zero. Let $\ell \geq 3$ be an integer, let $C_f > 0$, and let $0 < \delta_f \leq 1$. We call an entire function g with $g(0) = 1$ *admissible with these parameters* if

$$(2) \quad |g(z)| \leq \exp\{a_R(\operatorname{Re} z)^2 + a_I(\operatorname{Im} z)^2\} \quad (z \in \mathbb{C}),$$

$$(3) \quad |g(z) - 1| \leq C_f |z|^\ell \quad (|z| \leq \delta_f).$$

Put $f = g - 1$ and $a_{\max} = \max\{a_R, a_I\}$.

The two factors used later are $g_{\text{pm}}(z) = (1+z)e^{-z+z^2/2}$, with envelope weights $(a_R, a_I) = (1, 0)$ and vanishing order three, and $g_{\text{ls}}(z) = \cosh(z)e^{-z^2/2}$, with weights $(0, 1/2)$ and vanishing order four. Their precise bounds are stated in the corresponding applications and proved in Appendix C.

For a fixed scalar factor, the envelope matrix below is the quadratic form that bounds the absolute value of the full product. Its spectral margin is exactly what keeps that bound integrable. Notice that the condition is one-sided whenever one of the two envelope weights is zero.

Definition 2.2 (Admissible Gaussian moment matrix). Fix the envelope weights a_R and a_I . For a real symmetric matrix K , define the positive-semidefinite envelope matrix

$$K_{\text{env}} = a_R K_+ + a_I K_-.$$

Let $\beta \geq 0$ and $0 < d \leq 1$. We call a real symmetric n by n matrix K (β, d) -admissible if

$$(4) \quad \max_{i,j} |K_{ij}| \leq \frac{\beta}{n}, \quad I - 2K_{\text{env}} \succeq dI.$$

Given an admissible scalar factor g and a (β, d) -admissible matrix K of dimension n , retain the notation $f = g - 1$. For $S \subseteq [n]$, define

$$W_K(S) = \mathbb{E}_{K[S]} \prod_{i \in S} f(\phi_i), \quad W_K(\emptyset) = 1, \quad \mathcal{Z}_K = \mathbb{E}_K \prod_{i=1}^n g(\phi_i).$$

Theorem 2.3 (Diffuse Gaussian-product truncation). Fix $a_R, a_I, \ell, C_f, \delta_f, \beta$, and d as in Definitions 2.1 and 2.2, and set

$$\alpha = \frac{\ell}{2} - 1 \geq \frac{1}{2}.$$

There are positive constants $c_{\text{tr}}, a_{\text{tail}}, C_0$, and C_1 , and an integer n_0 , depending only on these fixed parameters, with the following properties. Suppose that g is admissible with the stated scalar parameters and that K is a (β, d) -admissible matrix of dimension n . Then all the expectations below converge absolutely and

$$(5) \quad \mathcal{Z}_K = \sum_{S \subseteq [n]} W_K(S).$$

If $n \geq n_0$ and R is an integer with $2 \leq R \leq c_{\text{tr}} n$, then

$$(6) \quad \left| \mathcal{Z}_K - \sum_{|S| < R} W_K(S) \right| \leq \left(C_1 \left(\frac{R}{n} \right)^\alpha \right)^R + e^{-a_{\text{tail}} n}.$$

Moreover,

$$(7) \quad |\mathcal{Z}_K - 1| \leq \frac{C_0}{n} \quad (n \geq n_0).$$

When $\beta > 0$, the constant c_{tr} may be chosen so that $c_{\text{tr}} \leq 1/(8\beta)$. Consequently, $\|K[S]\|_{\text{row}} \leq 1/8$ whenever $|S| \leq c_{\text{tr}}n$.

2.2. Scalar estimates, marginalization, and small coordinate sets. We prove the theorem in several steps. The following elementary remark records three consequences of the hypotheses on g .

Remark 2.4 (Useful scalar bounds). Admissibility immediately gives, for $0 < \delta \leq \delta_f$,

$$1 + |f(z)| \leq 3e^{a_R(\text{Re } z)^2 + a_I(\text{Im } z)^2}, \quad |f(z)| \leq C_f \delta^{\ell-2} |z|^2 \quad (|z| \leq \delta),$$

$$a_R(\text{Re}(\zeta w))^2 + a_I(\text{Im}(\zeta w))^2 \leq a_{\max} |\zeta|^2 |w|^2.$$

We use these three consequences without further comment.

The next lemma justifies restricting an indefinite complex Gaussian to a set of coordinates. Matching bilinear moments alone do not immediately give this conclusion for an arbitrary entire function, so we connect the two representations by analytic continuation.

Lemma 2.5 (Complex Gaussian marginalization). *Fix the envelope weights a_R and a_I . Let K be (β, d) -admissible, let $S \subseteq [n]$, and let $F : \mathbb{C}^S \rightarrow \mathbb{C}$ be entire. Suppose that*

$$|F(z)| \leq A \exp \left\{ \sum_{i \in S} (a_R(\text{Re } z_i)^2 + a_I(\text{Im } z_i)^2) \right\}$$

for some $A < \infty$. Then both expectations converge absolutely and

$$\mathbb{E}_K F((\phi_i)_{i \in S}) = \mathbb{E}_{K[S]} F(\phi).$$

The proof dilates both Gaussian representations by a complex parameter ζ . Interlacing transfers the spectral margin to $K[S]$, giving a common holomorphic neighborhood of $[0, 1]$, while Wick's rule identifies the Taylor series of the two representations at zero. The details are given in [Section C.3](#).

We next control a contribution whose support is smaller than a fixed fraction of all coordinates. The guiding count is simple. If a support has size s , an order- ℓ zero forces its Gaussian expansion to begin in degree ℓs . Pairing those legs costs roughly $1/n$ per edge, while the number of possible pairings contributes the corresponding power of s , up to constants depending only on ℓ . This suggests a factor $(s/n)^{\ell s/2}$. Complex dilation proves this estimate without enumerating the pairings.

Lemma 2.6 (Small supports). *Let g be an admissible scalar factor. Let $K = (K_{ij})$ be a real symmetric matrix of order n , and suppose that $\max_{i,j} |K_{ij}| \leq \beta/n$. If $\beta = 0$, then $K = 0$ and $W_K(S) = 0$ for every nonempty S . Suppose that $\beta > 0$, and put*

$$c_{\text{loc}} = \frac{1}{64a_{\max}\beta}, \quad B = 3(1 - 1/8)^{-1/2}, \quad C_{\text{loc}} = 16a_{\max}\beta B^{2/\ell}.$$

If S is nonempty and $s = |S| \leq c_{\text{loc}}n$, then the integral defining $W_K(S)$ converges absolutely and

$$(8) \quad |W_K(S)| \leq \left(C_{\text{loc}} \frac{s}{n} \right)^{\ell s/2}.$$

For a singleton, the stronger estimate

$$(9) \quad |W_K(\{i\})| \leq B(16a_{\max}\beta)^2 n^{-2}$$

holds.

Proof. The case $\beta = 0$ is immediate from the entrywise hypothesis. Assume $\beta > 0$. Put $L = K[S]$. Then $\|L\| \leq \|L\|_{\text{row}} \leq \beta s/n$. For a standard real Gaussian vector $x \in \mathbb{R}^s$, define

$$\Phi_S(\zeta) = \mathbb{E}_x \prod_{i \in S} f(\zeta(A_L x)_i), \quad \varrho^2 = \frac{n}{16a_{\max}\beta s}.$$

The assumption $s \leq c_{\text{loc}} n$ gives $\varrho \geq 2$. By Remark 2.4, for $|\zeta| \leq 3\varrho/2$,

$$\prod_{i \in S} |f(\zeta(A_L x)_i)| \leq 3^s \exp\{a_{\max} |\zeta|^2 x^\top |L| x\}.$$

Moreover, $2a_{\max}(3\varrho/2)^2 \|L\| \leq 9/32 < 1$. Thus the right-hand side gives a locally uniform integrable Gaussian majorant on $|\zeta| < 3\varrho/2$. In particular, Φ_S is holomorphic there and its value at $\zeta = 1$ converges absolutely.

On $|\zeta| = \varrho$, we have $2a_{\max}\varrho^2 \|L\| \leq 1/8$. Gaussian integration gives

$$|\Phi_S(\zeta)| \leq 3^s \det(I - 2a_{\max}\varrho^2 |L|)^{-1/2} \leq 3^s (1 - 1/8)^{-s/2} = B^s.$$

Condition (3), together with analyticity, says that f has a zero of order at least ℓ at the origin. Hence $\Phi_S(\zeta) = \zeta^{\ell s} H_S(\zeta)$ for a holomorphic function H_S . The maximum-modulus principle gives

$$|W_K(S)| = |\Phi_S(1)| \leq \varrho^{-\ell s} B^s = B^s \left(16a_{\max}\beta \frac{s}{n}\right)^{\ell s/2},$$

which is (8).

When $S = \{i\}$, the substitution $x \mapsto -x$ shows that Φ_S is even. Its order of vanishing is therefore an even integer at least ℓ , hence at least four. The same argument gives $|W_K(\{i\})| \leq B\varrho^{-4} = B(16a_{\max}\beta)^2 n^{-2}$. $\square$

2.3. Recombined large-support tail. The large-support estimate needs more information about small principal submatrices of the Hermitian covariance $|K|$ than a global operator norm provides.

Lemma 2.7 (Envelopes and principal submatrices). *Let K be a real symmetric n by n matrix, and let $\beta \geq 0$ satisfy $\max_{i,j} |K_{ij}| \leq \beta/n$. If $S \subseteq [n]$ has size s , then*

$$(10) \quad \| |K| [S] \| \leq \beta \sqrt{\frac{s}{n}},$$

$$(11) \quad |K|_{ii} \leq \frac{\beta}{\sqrt{n}}, \quad \text{tr } |K| [S] \leq \frac{\beta s}{\sqrt{n}}, \quad \text{tr } |K| \leq \beta \sqrt{n}.$$

Suppose in addition that g is admissible and K is (β, d) -admissible. For $T \subseteq [n]$, let Π_T be the coordinate projection, and put

$$B_T = a_R K_+^{1/2} \Pi_T K_+^{1/2} + a_I K_-^{1/2} \Pi_T K_-^{1/2}.$$

Then

$$(12) \quad \prod_{i \in T} |g(\phi_i)| \leq e^{x^\top B_T x}, \quad 0 \leq B_T \leq K_{\text{env}}, \quad \text{tr } B_T \leq a_{\max} \beta \sqrt{n}.$$

In particular, $I - 2B_T \succeq dI$ and

$$(13) \quad \mathbb{E}_x \prod_{i \in T} |g(\phi_i)| \leq \det(I - 2B_T)^{-1/2} \leq e^{C\sqrt{n}},$$

where C depends only on $(a_{\max}, \beta, d)$.

Proof. Let Π_S be the coordinate projection onto S . Since $|K|^2 = K^2$,

$$(\Pi_S |K| \Pi_S)^2 \preceq \Pi_S |K|^2 \Pi_S = \Pi_S K^2 \Pi_S.$$

Indeed, the difference is $\Pi_S |K| (I - \Pi_S) |K| \Pi_S \succeq 0$. Every entry of K^2 has modulus at most $\sum_{k=1}^n |K_{ik} K_{kj}| \leq \beta^2/n$. Consequently, $\|K^2[S]\| \leq \|K^2[S]\|_{\text{row}} \leq \beta^2 s/n$. Taking operator norms in the preceding positive-semidefinite inequality proves (10).

For every i , $|K|_{ii}^2 \leq (|K|^2)_{ii} = (K^2)_{ii} \leq \beta^2/n$. This gives the diagonal estimate. Summing it over $i \in S$, and then over all coordinates, gives the two trace estimates.

For the envelope assertion, sum (2) over $i \in T$. The real-coordinate contribution is $\sum_{i \in T} (\operatorname{Re} \phi_i)^2 = x^\top K_+^{1/2} \Pi_T K_+^{1/2} x$, and the imaginary contribution is analogous. This proves the first inequality in (12). Since $0 \preceq \Pi_T \preceq I$, we have $0 \preceq B_T \preceq K_{\text{env}}$. Moreover, $\operatorname{tr} B_T \leq \operatorname{tr} K_{\text{env}} \leq a_{\max} \operatorname{tr} |K| \leq a_{\max} \beta \sqrt{n}$ by (11).

The admissibility margin gives $I - 2B_T \succeq dI$. Gaussian integration yields the first inequality in (13). Finally, for $0 \preceq D \preceq (1-d)I$, the scalar inequality $-\log(1-u) \leq C_d u$ on $0 \leq u \leq 1-d$ gives $-\frac{1}{2} \log \det(I-D) \leq C_d \operatorname{tr} D$. Apply this with $D = 2B_T$ and use the trace bound above. $\square$

We now estimate the event on which many Gaussian coordinates are large. The required rate is stronger than an ordinary linear large-deviation bound. If $r = \rho n$, a gain $e^{-c\rho n}$ can lose to the entropy $e^{nh_2(\rho)}$, because $h_2(\rho)$ is of order $\rho \log(1/\rho)$ near zero. A witness-set estimate instead gives $e^{-c\sqrt{\rho}n}$, which wins since $\rho \log(1/\rho) = o(\sqrt{\rho})$. This square-root rate is the reason for working with small principal submatrices of $|K|$. For $1 \leq r \leq n$ and $\delta > 0$, let $\mathcal{A}_{r,\delta}$ be the event that at least r coordinates satisfy $|\phi_i| > \delta$. For a measurable event $\mathcal{B} \subseteq \mathbb{R}^n$, write

$$\mathcal{J}_T(\mathcal{B}) = \mathbb{E}_x \left[\mathbf{1}_{\{\mathcal{B}\}} \prod_{i \in T} |g(\phi_i)| \right].$$

Lemma 2.8 (Many large coordinates). *Let g be admissible in the sense of Definition 2.1. Suppose that K is (β, d) -admissible and $\beta > 0$. There is a constant C , depending only on $(a_{\max}, \beta, d)$, such that, uniformly over $T \subseteq [n]$,*

$$(14) \quad \mathcal{J}_T(\mathcal{A}_{r,\delta}) \leq \exp \left\{ -\frac{d\delta^2}{4\beta} \sqrt{rn} + nh_2(r/n) + C\sqrt{n} \right\}.$$

Proof. If $\mathcal{A}_{r,\delta}$ occurs, some set S of r coordinates satisfies $\sum_{i \in S} |\phi_i|^2 \geq r\delta^2$. For such a set, define

$$M_S = K_+^{1/2} \Pi_S K_+^{1/2} + K_-^{1/2} \Pi_S K_-^{1/2}.$$

The coordinate energy on S is $x^\top M_S x$. To estimate the norm of M_S , let $R_S : \mathbb{R}^n \rightarrow \mathbb{R}^r$ be coordinate restriction and put

$$C_S = \begin{pmatrix} R_S K_+^{1/2} \\ R_S K_-^{1/2} \end{pmatrix}.$$

Then $C_S^\top C_S = M_S$. Since $K_+^{1/2} K_-^{1/2} = 0$, the matrix $C_S C_S^\top$ is block diagonal with blocks $K_+[S]$ and $K_-[S]$. The nonzero eigenvalues of $C_S^\top C_S$ and $C_S C_S^\top$ agree. Because $K_+[S], K_-[S] \preceq |K|[S]$, Lemma 2.7 gives

$$(15) \quad \|M_S\| \leq \beta \sqrt{\frac{r}{n}}, \quad \operatorname{tr} M_S = \operatorname{tr} |K|[S] \leq \frac{\beta r}{\sqrt{n}}.$$

Fix one witness set S , and put $\lambda = d/(4\beta\sqrt{r/n})$. By Markov's inequality and Lemma 2.7, the part of $\mathcal{J}_T$ on which $x^\top M_S x \geq r\delta^2$ is at most

$$(16) \quad e^{-\lambda r \delta^2} \det(I - 2B_T - 2\lambda M_S)^{-1/2}.$$

Indeed, (15) gives $2\lambda \|M_S\| \leq d/2$, and $I - 2B_T \succeq dI$. Hence

$$I - 2B_T - 2\lambda M_S \succeq \frac{d}{2} I.$$

The determinant estimate from Lemma 2.7 applies to $2B_T + 2\lambda M_S$. Its trace is bounded by

$$2a_{\max} \beta \sqrt{n} + 2\lambda \frac{\beta r}{\sqrt{n}} = 2a_{\max} \beta \sqrt{n} + \frac{d}{2} \sqrt{r} \leq C\sqrt{n}.$$

Thus (16) is at most $\exp\{-d\delta^2 \sqrt{rn}/(4\beta) + C\sqrt{n}\}$. There are at most $\binom{n}{r} \leq \exp\{nh_2(r/n)\}$ witness sets. A union bound proves (14). $\square$

The small-subset bound cannot be summed all the way to size n . The next lemma instead recombines all subsets beyond one linear cutoff before taking an absolute value.

Lemma 2.9 (Recombined large-support tail). *Fix the parameters in [Definitions 2.1](#) and [2.2](#). Let g be admissible with the fixed scalar parameters, and let K be a (β, d) -admissible matrix of dimension n . For every $c_{\max} > 0$ there are constants $0 < c_* \leq c_{\max}$ and $a_{\text{tail}} > 0$, and an integer n_{tail} , depending only on the fixed parameters and $c_{\max}$, such that*

$$(17) \quad \left| \sum_{|S| \geq \lceil c_* n \rceil} W_K(S) \right| \leq e^{-a_{\text{tail}} n} \quad (n \geq n_{\text{tail}}).$$

Proof. If $\beta = 0$, then $K = 0$ and every nonempty contribution vanishes. We may therefore assume that $\beta > 0$.

Put $t_0 = d/(8(\beta + 1))$. Choose $0 < \delta \leq \delta_f$ so small that $\chi := C_f \delta^{\ell-2}/t_0$ satisfies $3\chi < 1$, and put $a_{\text{ld}} = d\delta^2/(4\beta)$. We shall choose $0 < c_* \leq \min\{c_{\max}, 1/2\}$ after obtaining the two estimates below. For a fixed positive c_* , define

$$s_* = \lceil c_* n \rceil, \quad r = \left\lceil \frac{s_*}{2} \right\rceil, \quad \mathcal{T}_{s_*}(\phi) = \sum_{|S| \geq s_*} \prod_{i \in S} f(\phi_i).$$

We take n large enough that $s_* \geq 2$.

First suppose that fewer than r coordinates are larger than δ in modulus. On this event, put $H = \{i : |\phi_i| > \delta\}$ and $A_{H^c} = \sum_{i \notin H} |f(\phi_i)|$. Then $|H| < r$, so every subset of size at least s_* uses at least $k = s_* - r$ coordinates outside H . If nonnegative numbers a_i are indexed outside H , their elementary symmetric polynomials satisfy

$$e_j((a_i)) \leq \frac{(\sum_i a_i)^j}{j!}, \quad \sum_{j \geq k} e_j((a_i)) \leq e^{\sum_i a_i} \frac{(\sum_i a_i)^k}{k!}.$$

The first inequality follows by expanding $(\sum_i a_i)^j$, and the second uses $k!/(k+j)! \leq 1/j!$ for $j \geq 0$.

Apply this estimate with $a_i = |f(\phi_i)|$. Summing first over the part of a selected set contained in H gives

$$(18) \quad |\mathcal{T}_{s_*}(\phi)| \leq \prod_{i \in H} (1 + |f(\phi_i)|) e^{A_{H^c}} \frac{A_{H^c}^k}{k!}.$$

By [Remark 2.4](#), $A_{H^c} \leq C_f \delta^{\ell-2} x^\top |K| x$. The same remark and $|H| < r$ give $\prod_{i \in H} (1 + |f(\phi_i)|) \leq 3^r e^{x^\top K_{\text{env}} x}$. Finally, if $Q = x^\top |K| x$, then $Q^k/k! \leq t_0^{-k} e^{t_0 Q}$. Substitution in (18) yields

$$(19) \quad |\mathcal{T}_{s_*}(\phi)| \leq 3^r \chi^{s_*-r} \exp \left\{ x^\top (K_{\text{env}} + (C_f \delta^{\ell-2} + t_0) |K|) x \right\}.$$

Since $C_f \delta^{\ell-2} < t_0/3$ and $\|K\| \leq \beta$,

$$2(C_f \delta^{\ell-2} + t_0) \|K\| \leq \frac{8}{3} t_0 \beta = \frac{d\beta}{3(\beta+1)} \leq \frac{d}{3}.$$

Together with $I - 2K_{\text{env}} \succeq dI$, this gives

$$I - 2K_{\text{env}} - 2(C_f \delta^{\ell-2} + t_0) |K| \succeq \frac{d}{2} I.$$

The trace of the positive-semidefinite matrix in the exponent of (19) is at most $(a_{\max} + C_f \delta^{\ell-2} + t_0) \text{tr} |K| \leq C\sqrt{n}$. Gaussian integration and the determinant estimate used in [Lemma 2.7](#) therefore give

$$(20) \quad \left| \mathbb{E}_x \left[\mathcal{T}_{s_*}(\phi) \mathbf{1}_{\{\mathcal{A}_{r,\delta}^c\}} \right] \right| \leq e^{C\sqrt{n}} 3^r \chi^{s_*-r}.$$

We next consider $\mathcal{A}_{r,\delta}$. The exact pointwise identity

$$(21) \quad \mathcal{T}_{s_*}(\phi) = \prod_{i=1}^n g(\phi_i) - \sum_{|S| < s_*} \prod_{i \in S} f(\phi_i)$$

preserves the cancellation among all subsets beyond the cutoff. For a fixed set S ,

$$\prod_{i \in S} |f(\phi_i)| \leq \prod_{i \in S} (1 + |g(\phi_i)|) = \sum_{T \subseteq S} \prod_{i \in T} |g(\phi_i)|.$$

Apply [Lemma 2.8](#) to each partial product on the right, and also to the full product in (21). Uniformity in T gives

$$(22) \quad \left| \mathbb{E}_x \left[\mathcal{T}_{s_*}(\phi) \mathbf{1}_{\{\mathcal{A}_{r,\delta}\}} \right] \right| \leq e^{-a_{\text{ld}} \sqrt{rn} + nh_2(r/n) + C\sqrt{n}} \left(1 + \sum_{s < s_*} \binom{n}{s} 2^s \right).$$

For fixed $0 < c_* \leq 1/2$, the standard entropy bound gives

$$\sum_{s < s_*} \binom{n}{s} 2^s \leq \exp\{n(h_2(c_*) + c_* \log 2) + o(n)\}.$$

All constants and the $o(n)$ term in the two preceding estimates are uniform over matrices satisfying the hypotheses. Since $r/n \rightarrow c_*/2$ and $(s_* - r)/n \rightarrow c_*/2$, put

$$\Psi(c) = -a_{\text{ld}} \sqrt{c/2} + h_2(c/2) + h_2(c) + c \log 2.$$

Combining (20) and (22) gives, uniformly over every admissible matrix sequence,

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log |\mathbb{E}_x \mathcal{T}_{s_*}(\phi)| \leq \max \left\{ \frac{c_*}{2} \log(3\chi), \Psi(c_*) \right\}.$$

The first quantity is negative because $3\chi < 1$. The positive terms in the second quantity are $O(c_* \log(1/c_*))$, which is $o(\sqrt{c_*})$ as $c_* \downarrow 0$. We may therefore choose $0 < c_* \leq \min\{c_{\max}, 1/2\}$ so that both quantities are strictly negative. Taking half of the smaller resulting margin and increasing n_{tail} proves

$$|\mathbb{E}_x \mathcal{T}_{s_*}(\phi)| \leq e^{-a_{\text{tail}} n}.$$

Finally, expand $\mathcal{T}_{s_*}$ into its finite sum. Applying [Lemma 2.5](#) to each term shows that

$$\mathbb{E}_x \mathcal{T}_{s_*}(\phi) = \sum_{|S| \geq s_*} W_K(S).$$

This is (17). □

2.4. From tail bounds to an efficient cutoff.

Proof of Theorem 2.3. If $\beta = 0$, then the entrywise hypothesis forces $K = 0$. Consequently, $\mathcal{Z}_K = 1$ and every nonempty $W_K(S)$ vanishes, so all conclusions are immediate. Assume from now on that $\beta > 0$.

By [Lemma 2.7](#) with $T = [n]$, the expectation defining $\mathcal{Z}_K$ converges absolutely. Expanding $g = 1 + f$ gives a finite pointwise sum over coordinate subsets. For each set S , [Remark 2.4](#) gives the envelope required by [Lemma 2.5](#) for $\prod_{i \in S} f(\phi_i)$. Marginalizing every term proves (5), as well as absolute convergence of every $W_K(S)$.

Put $C_2 = eC_{\text{loc}}^{\ell/2}$. For $2 \leq s \leq c_{\text{loc}} n$, [Lemma 2.6](#) and $\binom{n}{s} \leq (en/s)^s$ give

$$(23) \quad \sum_{|S|=s} |W_K(S)| \leq \Lambda_s, \quad \Lambda_s = \left(C_2 \left(\frac{s}{n} \right)^{\alpha} \right)^s.$$

Moreover,

$$\frac{\Lambda_{s+1}}{\Lambda_s} = C_2 \left(\frac{s+1}{n} \right)^\alpha \left(1 + \frac{1}{s} \right)^{\alpha s} \leq C_2 e^\alpha \left(\frac{s+1}{n} \right)^\alpha.$$

Choose $c_{\max} > 0$ so small that

$$(24) \quad c_{\max} \leq c_{\text{loc}}, \quad \beta c_{\max} \leq \frac{1}{8}, \quad C_2 e^\alpha (2c_{\max})^\alpha \leq \frac{1}{2}.$$

After increasing n_0 , we may assume $c_{\max}n \geq 1$. If $s < c_{\max}n$, then $(s+1)/n \leq 2c_{\max}$, so the layer bounds decrease by a factor of at least two throughout this range.

Apply [Lemma 2.9](#) with this $c_{\max}$, and set $c_{\text{tr}} = c_*$. Increase n_0 to include the threshold in that lemma. Let $2 \leq R \leq c_{\text{tr}}n$, and put $s_* = \lceil c_{\text{tr}}n \rceil$. The omitted subsets split into the layers $R \leq s < s_*$ and the signed tail beginning at s_* . By (23) and (24), the absolute values of the intermediate layers form a geometric series bounded by $2\Lambda_R$. The linear-cut lemma controls the remaining tail. Hence

$$\left| \mathcal{Z}_K - \sum_{|S| < R} W_K(S) \right| \leq 2 \left(C_2 \left(\frac{R}{n} \right)^\alpha \right)^R + e^{-a_{\text{tail}}n}.$$

This proves (6) with $C_1 = \sqrt{2}C_2$. We choose the separate constant C_0 for the near-one estimate below.

For (7), the singleton bound gives $\sum_{i=1}^n |W_K(\{i\})| \leq B(16a_{\max}\beta)^2/n$. The layers beginning at size two contribute at most $2\Lambda_2 = O(n^{-2\alpha}) = O(n^{-1})$, and the signed linear tail is exponentially small. Enlarging C_0 proves (7). Finally, $c_{\text{tr}} \leq c_{\max} \leq 1/(8\beta)$, and $\|K[S]\|_{\text{row}} \leq \beta|S|/n \leq 1/8$ when $|S| \leq c_{\text{tr}}n$. $\square$

The truncation bound becomes algorithmic because its cutoff can be chosen without paying $n^{O(\log(1/\varepsilon))}$ time. We record the elementary calculus fact once.

Lemma 2.10 (Least truncation cutoff). *Fix $\alpha \geq 1/2$, $C_2 > 0$, and $c_{\text{tr}} > 0$. There is a constant $\eta > 0$, depending only on $(\alpha, C_2, c_{\text{tr}})$, with the following property. Let $b \geq 1$ and suppose that $b < \eta n$. The least integer $R \geq 2$ satisfying*

$$(25) \quad \left(C_2 \left(\frac{R}{n} \right)^\alpha \right)^R \leq 2^{-b}$$

exists, satisfies $R \leq c_{\text{tr}}n$, and obeys

$$(26) \quad R \log \frac{en}{R} = O(b + \log n).$$

The hidden constant depends only on $(\alpha, C_2, c_{\text{tr}})$.

Writing $D = C_2^{1/\alpha}$, the cutoff condition is governed by $F(x) = x \log(n/(Dx))$. Monotonicity and minimality give $F(R) = O(b + \log n)$ and hence (26). The elementary details are given in [Section C.5](#).

3. PERFECT MATCHINGS IN DENSE GRAPHS

3.1. The matrix theorem and its consequences. This section states the matrix-level matching theorem used by the algorithm, two scaling criteria that reduce weighted matching counts to it, and representative consequences. Its proof combines the common truncation principle of [Section 2](#) with the matching-specific normalization and integral representation developed below. The longer deductions from the scaling criteria are deferred to the appendices.

3.1.1. *The matrix theorem.* For a positive integer k , let $\mathbf{J}_k$, I_k , and $\mathbf{1}_k$ denote the all-ones matrix, the identity matrix, and the all-ones column vector of dimension k , respectively. We omit subscripts when the dimension is clear. Throughout, $\|\cdot\|$ and $\|\cdot\|_F$ denote the spectral operator and Frobenius norms.

For even n , define the stochastic complete-graph reference matrix

$$P_0 = \frac{\mathbf{J}_n - I_n}{n-1}.$$

In the balanced bipartite case, write $n = 2m$ and define

$$P_b = \begin{pmatrix} 0 & \mathbf{J}_m/m \\ \mathbf{J}_m/m & 0 \end{pmatrix}.$$

Both reference matrices are symmetric and stochastic with zero diagonal. The following theorem says that a diffuse centered perturbation of either matrix admits a deterministic relative approximation.

Theorem 3.1 (Hafnian approximation for centered perturbations). *Fix $0 \leq \beta < \infty$ and $\kappa > 0$. For either reference matrix P above, let n be even, let $0 < \varepsilon < 1$, and let E be a real symmetric n by n matrix. Suppose that $P + E$ is entrywise nonnegative and*

$$(27) \quad E\mathbf{1} = 0, \quad E_{ii} = 0, \quad \max_{i,j} |E_{ij}| \leq \frac{\beta}{n}, \quad \|E\| \leq 1 - \kappa.$$

In the complete-bipartite case we additionally require $n = 2m$ and

$$E = \begin{pmatrix} 0 & D \\ D^\top & 0 \end{pmatrix}$$

for an m by m matrix D .

There is a deterministic real-arithmetic algorithm that returns a number $\hat{H} \geq 0$ satisfying

$$e^{-\varepsilon} \text{haf}(P + E) \leq \hat{H} \leq e^{\varepsilon} \text{haf}(P + E).$$

It uses

$$n^{O_{\beta,\kappa}(1)} \varepsilon^{-O_{\beta,\kappa}(1)}$$

arithmetic and elementary-function operations. The implicit constants depend only on β and κ .

In the bipartite case, the block condition ensures that every matching selected from E is bipartite. The coefficient contributed by the reference matrix after this selection depends only on the matching size. This is the additional structural input needed for the inverse-Gamma identity. In fact, (43) proves that $\text{haf}(P + E) > 0$ for all sufficiently large n under these conditions. For such n , entrywise nonnegativity may be omitted: the same procedure returns $\hat{H}$ satisfying $|\hat{H} - \text{haf}(P + E)| \leq (e^{\varepsilon} - 1) |\text{haf}(P + E)|$. We retain nonnegativity in the all-dimensions statement so that multiplicative approximation has its usual meaning in the finitely many smaller dimensions handled by exact evaluation.

Remark 3.2 (Doubly stochastic inputs). The argument of Theorem 3.1, with the finite-precision analysis in Appendix D, gives a deterministic FPTAS for $\text{per}(Y)$ when Y is a rational doubly stochastic m by m matrix satisfying $\max_{i,j} Y_{ij} \leq C/m$ and $\sigma_2(Y) \leq 1 - \kappa$, where $\sigma_2(Y)$ is the second-largest singular value of Y . Its bit complexity is $m^{O_{C,\kappa}(1)} \varepsilon^{-O_{C,\kappa}(1)} \text{poly}(L_Y + L_{\varepsilon})$. Indeed, apply the theorem to the symmetric dilation of $Y - \mathbf{J}_m/m$. The entrywise hypothesis is independent of the spectral one: for fixed $0 < a < 1$, the matrix $aI + (1 - a)\mathbf{J}_m/m$ has second singular value a but diagonal entries of order one. On the diffuse class above, the conclusion strengthens McCullagh's prescribed $O(1/m)$ relative error to arbitrary requested accuracy [McC14].

3.1.2. *Entropy-scaling criteria.* For a graph G , let A_G and D_G be its adjacency and degree matrices, and put $L_G = D_G - A_G$ and $Q_G = D_G + A_G$. Write $\lambda_2(L_G)$ for the second-smallest eigenvalue of L_G . When $\mathcal{D}(G)$ contains a point positive on every edge, let X denote the matrix formed from the maximizer in the definition of $h_A(G)$. The normalization in [Proposition 3.9](#) shows that $X_{uv} = a_{uv}r_ur_v$ on the support.

A nonnegative square matrix has *total support* if its bipartite support has a perfect matching and every positive entry belongs to one. For such a matrix, let $Y_{ij}^* = b_{ij}r_ic_j$ denote its weighted Sinkhorn scaling, with the gauge $\prod_i r_i = \prod_j c_j$. We write $\sigma_2(Y^*)$ for its second-largest singular value.

The symmetric criterion below assumes only that $\mathcal{D}(G)$ contains a point positive on every edge. This condition holds whenever every edge of G belongs to a perfect matching. To apply the matrix theorem to an input A , it remains to know when entropy scaling produces a perturbation satisfying its entrywise and spectral hypotheses. The next criterion expresses this in terms of the scaling factors and the Laplacian and signless Laplacian of the support. Both criteria are promise results: the algorithms are guaranteed on inputs satisfying the displayed hypotheses and need not verify those hypotheses.

Theorem 3.3 (Symmetric scaling criterion). *Fix $0 < \theta \leq 1$ and positive constants $c_{\text{lo}}, c_{\text{hi}}, q', q$. Let A be a rational symmetric n by n matrix, where n is even, with zero diagonal and entries in $\{0\} \cup [\theta, 1]$. Suppose that its support graph G has the property that $\mathcal{D}(G)$ contains a point positive on every edge. Write the maximum-entropy scaling as $X_{uv} = a_{uv}r_ur_v$ on the edges, and suppose that*

$$\frac{c_{\text{lo}}}{\sqrt{n}} \leq r_v \leq \frac{c_{\text{hi}}}{\sqrt{n}} \quad (v \in V(G)), \quad \lambda_2(L_G) \geq q'n, \quad Q_G \succeq qnI.$$

Then, given $0 < \varepsilon < 1$, a deterministic algorithm returns $\hat{H} \geq 0$ satisfying

$$e^{-\varepsilon} \text{haf}(A) \leq \hat{H} \leq e^{\varepsilon} \text{haf}(A).$$

If G has a perfect matching, then $\hat{H} > 0$. Its bit complexity is

$$n^{O_{\theta, c_{\text{lo}}, c_{\text{hi}}, q', q}(1)} \varepsilon^{-O_{\theta, c_{\text{lo}}, c_{\text{hi}}, q', q}(1)} \text{poly}(L_A + L_{\varepsilon}).$$

For all sufficiently large n , the hypotheses of [Theorem 3.3](#) themselves force the support to contain a perfect matching. See [Corollary 3.20](#).

The bipartite analogue has one fewer spectral hypothesis. The singular-value gap supplies both the operator-norm bound needed by the matrix theorem and the strong convexity needed to compute the Sinkhorn scaling.

Theorem 3.4 (Bipartite scaling criterion). *Fix $0 < \theta \leq 1$, constants $0 < c_{\text{lo}} \leq c_{\text{hi}} < \infty$, and $0 < \kappa_{\text{b}} \leq 1$. Let B be a rational m by m matrix with entries in $\{0\} \cup [\theta, 1]$ and total support. Let $Y_{ij}^* = b_{ij}r_ic_j$ be its Sinkhorn scaling, and choose the factors in the gauge $\prod_i r_i = \prod_j c_j$. Suppose that*

$$\frac{c_{\text{lo}}}{\sqrt{m}} \leq r_i, c_j \leq \frac{c_{\text{hi}}}{\sqrt{m}}, \quad \sigma_2(Y^*) \leq 1 - \kappa_{\text{b}}.$$

Then, given $0 < \varepsilon < 1$, a deterministic algorithm returns $\hat{Z}_B > 0$ satisfying

$$e^{-\varepsilon} \text{per}(B) \leq \hat{Z}_B \leq e^{\varepsilon} \text{per}(B).$$

Its bit complexity is

$$m^{O_{\theta, c_{\text{lo}}, c_{\text{hi}}, \kappa_{\text{b}}}(1)} \varepsilon^{-O_{\theta, c_{\text{lo}}, c_{\text{hi}}, \kappa_{\text{b}}}(1)} \text{poly}(L_B + L_{\varepsilon}).$$

The symmetric criterion is proved in [Section 3.2](#), after the normalization has been constructed. The bipartite criterion is proved in [Appendix B](#).

3.1.3. Consequences of the scaling criteria. We first record the regular cases, where the stochastic normalization is already visible in the input. The irregular consequences that follow illustrate the role of entropy scaling. Proofs of the corollaries in this subsection are given in [Appendix A](#) and [Appendix B](#).

Corollary 3.5 (Dense regular spectral expanders). *Fix $0 < c < 1$ and $0 < \kappa < 1$.*

- (a) *Let G be a d -regular graph on an even number n of vertices, where $d \geq cn$, and suppose that $\|A_G z\|_2 \leq (1 - \kappa)d\|z\|_2$ for every $z \perp \mathbf{1}$. There is a deterministic FPTAS for $\# \text{PM}(G)$, with running time $n^{O_{c,\kappa}(1)} \varepsilon^{-O_{c,\kappa}(1)}$.*
- (b) *Let B be the zero-one biadjacency matrix of a d -regular bipartite graph with m vertices on each side, where $d \geq cm$, and suppose that $\sigma_2(B/d) \leq 1 - \kappa$. There is a deterministic FPTAS for $\text{per}(B)$, with running time $m^{O_{c,\kappa}(1)} \varepsilon^{-O_{c,\kappa}(1)}$.*

In both parts the input is already scaled. Ebrahimnejad, Nagda, and Oveis Gharan obtain randomized approximate counting and sampling in polynomial time for regular strong expanders [\[ENOG22\]](#). Gamarnik and Katz obtain a deterministic $(1 + \eta)^m$ -factor approximation for constant-degree bipartite expanders [\[GK10\]](#). The corollary above is deterministic and gives arbitrary relative accuracy, at the price of a degree linear in the order and a two-sided spectral gap.

Whether every edge of a graph belongs to a perfect matching can be checked in polynomial time: for each edge uv , one tests whether $G - \{u, v\}$ has a perfect matching using Edmonds' algorithm [\[Edm65\]](#). The next two consequences are the main applications of entropy scaling beyond the fixed-margin Dirac class. They handle irregular supports whose density may lie below one half.

Corollary 3.6 (Dense supports with degree and codegree bounds). *Fix constants $p > 0$, $0 \leq \rho < 1$, $\eta > 0$, and $q > 0$ such that $\eta > 2\rho p$. There is a deterministic FPTAS for $\# \text{PM}(G)$ on every even-order graph G in which every edge belongs to a perfect matching and which satisfies the following conditions, where $d_G(v)$ denotes the degree of a vertex v :*

$$\begin{aligned} (1 - \rho)pn &\leq d_G(v) \leq (1 + \rho)pn && \text{for every vertex } v, \\ |N(u) \cap N(v)| &\geq \eta n && \text{for all distinct } u, v, \\ Q_G &\succeq qnI, \end{aligned}$$

where Q_G is the signless Laplacian. Its running time is $n^{O_{p,\rho,\eta,q}(1)} \varepsilon^{-O_{p,\rho,\eta,q}(1)}$. For every fixed $p \in (0, 1)$, suitable constants $\rho, \eta, q > 0$ make these hypotheses hold with probability tending to one for the Erdős–Rényi graph $G(n, p)$ along even values of n .

The condition $Q_G \succeq qnI$ is a quantitative nonbipartiteness hypothesis. The common-neighbor condition makes G connected. For a connected graph, Q_G is singular exactly when G is bipartite. This is why part (b) of [Theorem 1.1](#) uses the separate reference matrix P_b rather than following from [Corollary 3.6](#).

The bipartite formulation factors out this unavoidable -1 mode and therefore requires no additional signless-Laplacian hypothesis. The following consequence reaches zero-one matrices at every fixed positive density.

Corollary 3.7 (Bipartite supports with degree and codegree bounds). *Fix $0 < \theta \leq 1$, $p > 0$, $0 \leq \rho < 1$, and $\eta > 0$. Put $d_- = (1 - \rho)p$, $d_+ = (1 + \rho)p$, and suppose that*

$$(28) \quad \eta > d_+ - \theta d_-.$$

Let B be a rational m by m matrix with entries in $\{0\} \cup [\theta, 1]$ and total support. Suppose that every row and column of its support has degree between d_-m and d_+m , and that every two distinct rows, and every two distinct columns, have at least ηm common neighbors. Then there is a deterministic FPTAS for $\text{per}(B)$, with bit complexity

$$m^{O_{p,\rho,\eta,\theta}(1)} \varepsilon^{-O_{p,\rho,\eta,\theta}(1)} \text{poly}(L_B + L_\varepsilon).$$

In particular, if B is a zero-one matrix with independent Bernoulli(p) entries for any fixed $p \in (0, 1]$, then its support satisfies these hypotheses with probability tending to one as $m \rightarrow \infty$.

The proof, including the random-support assertion, is given in [Appendix B](#).

3.2. Maximum-entropy normalization. We now turn from the range of applications to the proof. This section factors the leading entropy out of each input and verifies the centered, entrywise, and spectral hypotheses of [Theorem 3.1](#). The symmetric and bipartite constructions differ only at this normalization stage. After the symmetric dilation below, the original support graphs and weight matrices no longer enter the analytic argument.

3.2.1. The symmetric normalization. The first step removes the leading weighted entropy of the input matrix. For an adjacency matrix, this is the entropy parameter used by Cuckler and Kahn to determine the exponential scale of the count [\[CK09a, CK09b\]](#). Here it also supplies an exact diagonal normalization. Let $A = (a_{uv})$ satisfy part (a) of [Theorem 1.1](#), let G be its support graph, and recall the set $\mathcal{D}(G)$ defined in the introduction. Only the degree equations appear in $\mathcal{D}(G)$. In particular, this is not the perfect-matching polytope with its odd-cut inequalities. We assume in this section that $n \geq 1/\gamma + 2$. The finitely many smaller inputs are assigned to the exact branch.

Lemma 3.8 (Every edge extends to a perfect matching). *Suppose that [\(1\)](#) holds and that $n \geq 1/\gamma + 2$. Every edge of G belongs to a perfect matching.*

Proof. Fix an edge uv . To extend uv , it is enough to find a perfect matching after deleting its endpoints. The remaining graph has $n - 2$ vertices and minimum degree at least $(\frac{1}{2} + \gamma)n - 2 \geq \frac{n-2}{2}$. Dirac's theorem gives a Hamiltonian cycle in this graph. Alternating edges of the cycle form a perfect matching, which extends to one of G after adding uv . $\square$

For a vector $w \in \mathcal{D}(G)$, define its weighted entropy by $F_A(w) = \sum_{uv \in E(G)} w_{uv} \log \frac{a_{uv}}{w_{uv}}$, where a summand with $w_{uv} = 0$ is interpreted as zero. The set $\mathcal{D}(G)$ is nonempty because G has a perfect matching. Let w^* maximize F_A , so $F_A(w^*) = h_A(G)$. By [Lemma 3.8](#), averaging one perfect matching through each edge gives a point of $\mathcal{D}(G)$ that is positive on every edge.

Proposition 3.9 (Exact entropy normalization). *Let A be any nonnegative symmetric matrix with zero diagonal and support graph G . Suppose that $\mathcal{D}(G)$ contains a point positive on every edge, and let w^* maximize F_A on $\mathcal{D}(G)$. Then w^* is the unique maximizer and is positive on every edge. There are positive numbers r_v , indexed by the vertices of G , such that*

$$w_{uv}^* = a_{uv} r_u r_v \quad (uv \in E(G)).$$

Let X be the symmetric matrix with entries $X_{uv} = w_{uv}^$ on edges of G , and zero entries on nonedges and on the diagonal. Then*

$$(29) \quad \text{haf}(A) = e^{h_A(G)} \text{haf}(X).$$

Proof. For every $a > 0$, the function $x \mapsto x \log(a/x)$, extended continuously by zero at $x = 0$, is strictly concave on $[0, \infty)$. Thus F_A is strictly concave on the convex set $\mathcal{D}(G)$, and its maximizer is unique.

Let $\bar{w} \in \mathcal{D}(G)$ be positive on every edge, and suppose for a contradiction that w^* has a zero coordinate. For $w(t) = (1 - t)w^* + t\bar{w}$,

$$F_A(w(t)) - F_A(w^*) = t \log(1/t) \sum_{e: w_e^* = 0} \bar{w}_e + O(t) > 0$$

for all sufficiently small t . The displayed leading term comes from the zero coordinates, while the positive coordinates change by $O(t)$. This contradiction proves that w^* is positive on every edge.

Let d_v be the multiplier for the degree equation at v . Stationarity gives $\log(w_{uv}^*/a_{uv}) = d_u + d_v - 1$. Hence, with $r_v = e^{d_v - 1/2}$, one has $w_{uv}^* = a_{uv}r_ur_v$. Moreover,

$$\sum_v \log r_v = \sum_{uv \in E(G)} w_{uv}^* (\log r_u + \log r_v) = \sum_{uv \in E(G)} w_{uv}^* \log \frac{w_{uv}^*}{a_{uv}} = -h_A(G).$$

Every perfect matching M uses each factor r_v once, so $\prod_{uv \in M} a_{uv} = (\prod_v r_v)^{-1} \prod_{uv \in M} X_{uv}$. Summing over M gives $\text{haf}(A) = (\prod_v r_v)^{-1} \text{haf}(X) = e^{h_A(G)} \text{haf}(X)$, which is (29). $\square$

The degree margin forces every scaling factor to have order $n^{-1/2}$ and every nonzero entry of X to have order n^{-1} . Put $\alpha = 1/2 + \gamma$. For vertex sets S, T , write $X(S, T) = \sum_{u \in S, v \in T} X_{uv}$, and put $r(S) = \sum_{v \in S} r_v$.

Lemma 3.10 (Uniform bounds for the symmetric scaling). *The scaling factors satisfy*

$$(30) \quad \theta \sqrt{\frac{2\gamma}{n}} \leq r_v \leq \frac{1}{\theta \sqrt{2\gamma n}} \quad (v \in V(G)).$$

Consequently, every edge uv satisfies

$$(31) \quad \frac{2\gamma\theta^3}{n} \leq X_{uv} \leq \frac{1}{2\gamma\theta^2 n}.$$

Proof. We first show that the vertices with smaller scaling factors carry substantial total mass. Every neighborhood must meet that mass, which gives the upper factor bound. The row equations then give the lower one.

Write $N(v)$ for the neighborhood of a vertex v . Let V_{high} consist of the $\lfloor (1 - \alpha)n \rfloor$ vertices with largest r -values, and put $V_{\text{low}} = V(G) \setminus V_{\text{high}}$. Symmetry and stochasticity give $X(V_{\text{low}}, V_{\text{low}}) \geq |V_{\text{low}}| - |V_{\text{high}}| \geq 2\gamma n$. Indeed, the mass from V_{low} to V_{high} is at most the total column mass $|V_{\text{high}}|$. On the other hand, $a_{uv} \leq 1$ gives $X(V_{\text{low}}, V_{\text{low}}) \leq r(V_{\text{low}})^2$. Hence $r(V_{\text{low}}) \geq \sqrt{2\gamma n}$.

For every vertex v , the set $V(G) \setminus N(v)$ has at most $|V_{\text{high}}|$ vertices. Since V_{high} contains the largest factors, $\sum_{u \in N(v)} r_u = r(V(G)) - r(V(G) \setminus N(v)) \geq r(V(G)) - r(V_{\text{high}}) = r(V_{\text{low}})$. The row equation and $a_{uv} \geq \theta$ on the support now give $1 = r_v \sum_{u \in N(v)} a_{uv} r_u \geq \theta r_v \sqrt{2\gamma n}$, so $r_v \leq (\theta \sqrt{2\gamma n})^{-1}$. Conversely, the same row equation and $a_{uv} \leq 1$ give, with $r_{\text{max}} = \max_{u \in V(G)} r_u$, $1 \leq r_v(n-1)r_{\text{max}}$, and therefore $r_v \geq \frac{\theta \sqrt{2\gamma n}}{n-1} \geq \theta \sqrt{\frac{2\gamma}{n}}$. Since $X_{uv} = a_{uv}r_ur_v$ on an edge, these two factor bounds prove (31). $\square$

We next separate the constant mode. Recall the complete-graph reference matrix P_0 from Section 3.1, and put $E = X - P_0$. Both X and P_0 are stochastic, so $E\mathbf{1} = 0$. They also have zero diagonal.

To control E spectrally, we use the Laplacian L_G and signless Laplacian Q_G introduced before the scaling criteria. The Laplacian controls fluctuations orthogonal to $\mathbf{1}$, while the signless Laplacian rules out an almost-bipartite negative mode.

Lemma 3.11 (Spectral bounds for the support graph). *Suppose that (1) holds. Then*

$$\lambda_2(L_G) \geq 2\gamma n, \quad Q_G \succeq 2\gamma n I.$$

Proof. Let $H = \overline{G}$ be the complement of G , and write $\Delta(H)$ for its maximum degree. We have $\Delta(H) \leq (1/2 - \gamma)n - 1$. On $\mathbf{1}^\perp$, the identity $L_G = L_{K_n} - L_H$ and the bound $\lambda_{\text{max}}(L_H) \leq 2\Delta(H)$ give $\lambda_2(L_G) \geq n - 2\Delta(H) \geq 2\gamma n$. Likewise, $Q_G + Q_H = (n-2)I + \mathbf{J}$ and $\lambda_{\text{max}}(Q_H) \leq 2\Delta(H)$. Hence $Q_G \succeq ((n-2) - 2\Delta(H))I \succeq 2\gamma n I$. $\square$

3.2.2. The bipartite normalization. The permanent case uses the same analytic theorem after a different normalization. Weighted Sinkhorn scaling first removes the entropy, and a symmetric dilation then represents the permanent of the scaled matrix as a hafnian. We state the estimates needed for this reduction here and defer their elementary but longer proofs to [Appendix B](#).

Let $B = (b_{ij})$ be a nonnegative m by m matrix. Define $\mathcal{D}_{\text{bi}}(B)$ to be the polytope of doubly stochastic matrices whose nonzero entries occur only where B is positive:

$$\mathcal{D}_{\text{bi}}(B) = \left\{ Y \in \mathbb{R}_{\geq 0}^{m \times m} : Y_{ij} = 0 \text{ if } b_{ij} = 0, \quad Y\mathbf{1} = \mathbf{1}, \quad Y^\top \mathbf{1} = \mathbf{1} \right\}.$$

For $Y \in \mathcal{D}_{\text{bi}}(B)$, define $F_B(Y) = \sum_{i,j: b_{ij} > 0} Y_{ij} \log(b_{ij}/Y_{ij})$, with a zero summand when $Y_{ij} = 0$. When $\mathcal{D}_{\text{bi}}(B)$ is nonempty, write $h_B = \max_{Y \in \mathcal{D}_{\text{bi}}(B)} F_B(Y)$.

Recall that total support means that the bipartite support has a perfect matching and every positive entry belongs to one.

Proposition 3.12 (Bipartite entropy normalization). *Suppose that B has total support. Then the weighted entropy maximizer Y^* in $\mathcal{D}_{\text{bi}}(B)$ is positive at every positive entry and has the form $Y_{ij}^* = b_{ij} r_i c_j$ ($b_{ij} > 0$) for positive row factors r_i and column factors c_j . Moreover,*

$$(32) \quad \text{per}(B) = e^{h_B} \text{per}(Y^*).$$

The factorization follows from the same relative-interior and Lagrange-multiplier argument used in [Proposition 3.9](#), with separate multipliers for rows and columns. Existence and uniqueness of the doubly stochastic scaled matrix under total support are due to Sinkhorn and Knopp [\[SK67\]](#). See [Appendix B](#). The fixed degree margin gives the additional quantitative estimates needed below.

We now impose the degree-margin hypothesis in part (b) of [Theorem 1.1](#). The factors are unchanged if all r_i are multiplied by one positive constant and all c_j are divided by the same constant. We remove this ambiguity by requiring $\prod_i r_i = \prod_j c_j$. Equivalently, there are vectors x^* and y^* with $\sum_i x_i^* = \sum_j y_j^*$ such that $r_i = m^{-1/2} e^{x_i^*}$ and $c_j = m^{-1/2} e^{y_j^*}$.

Proposition 3.13 (Uniform bounds for the bipartite scaling). *Suppose that every row and column of the support of B contains at least $(1/2 + \gamma)m$ entries, that its nonzero entries lie in $[\theta, 1]$, and that $m \geq \lceil (2\gamma)^{-1} \rceil$. Then B has total support. Every positive entry satisfies*

$$(33) \quad \frac{2\gamma\theta^3}{m} \leq Y_{ij}^* \leq \frac{1}{2\gamma\theta^2 m}.$$

Under this normalization, there are positive constants $c_{\gamma,\theta}$ and $C_{\gamma,\theta}$ such that the individual factors satisfy

$$(34) \quad \frac{c_{\gamma,\theta}}{\sqrt{m}} \leq r_i, c_j \leq \frac{C_{\gamma,\theta}}{\sqrt{m}}.$$

Finally, let $\sigma_2(Y^)$ denote the second-largest singular value of Y^* . Then*

$$(35) \quad \sigma_2(Y^*) \leq 1 - 4\gamma^3\theta^6.$$

The proof combines Hall's theorem, a mass comparison for the row and column factors, and a common-neighbor estimate for the second singular value. See [Appendix B](#). Only the entrywise bounds, the uniform bounds on the factors, and the singular-value gap will be used in the main argument.

To apply the matrix theorem, represent the permanent as a hafnian by symmetric dilation. Recall P_b from [Section 3.1](#), and put $D = Y^* - \mathbf{J}_m/m$ and

$$E_b = \begin{pmatrix} 0 & D \\ D^\top & 0 \end{pmatrix}.$$

Then $E_b \mathbf{1} = 0$, its diagonal blocks vanish, and, in the ambient dimension $n = 2m$, one has $\max_{i,j} |(E_b)_{ij}| \leq (\gamma\theta^2 n)^{-1}$ and $\|E_b\| \leq 1 - 4\gamma^3\theta^6$. Indeed, the entrywise estimate follows from

(33), while $\|Y^* - \mathbf{J}_m/m\| = \sigma_2(Y^*)$ and the symmetric dilation has the same norm. Moreover, $\text{haf}(P_b + E_b) = \text{per}(Y^*)$ and $\text{haf}(P_b) = m!/m^m$.

Thus the bounds on the factors control the entries of E_b , while the singular-value gap controls its operator norm. The hypotheses of the matrix theorem follow. This dilation is the only step in which the bipartite reduction differs from the symmetric one.

3.2.3. Proof of the scaling criteria. The normalizations above give the identities and quantitative bounds used in the two criteria stated in [Section 3.1](#). We now verify their remaining hypotheses.

Proof of Theorem 3.3. It suffices to verify the hypotheses of the matrix theorem after the exact entropy normalization. The bounds on the factors give the entrywise bound, while the Laplacian and signless Laplacian keep the spectrum of $E = X - P_0$ away from $+1$ and -1 , respectively.

The support hypothesis allows us to apply [Proposition 3.9](#). The factor bounds give $\theta c_{\text{lo}}^2/n \leq X_{uv} \leq c_{\text{hi}}^2/n$ for $uv \in E(G)$. For $z \perp \mathbf{1}$, comparison with the two unweighted quadratic forms gives

$$z^\top (I - X)z \geq \frac{\theta c_{\text{lo}}^2}{n} z^\top L_G z, \quad z^\top (I + X)z \geq \frac{\theta c_{\text{lo}}^2}{n} z^\top Q_G z.$$

Thus the spectrum of X on $\mathbf{1}^\perp$ lies in $[-1 + \theta c_{\text{lo}}^2 q, 1 - \theta c_{\text{lo}}^2 q']$. Put $E = X - P_0$. Choose fixed constants satisfying $\beta > c_{\text{hi}}^2 + 2$ and $0 < \kappa < \min\{1/2, (\theta c_{\text{lo}}^2/2) \min\{q, q'\}\}$, and enlarge a fixed threshold n_0 so that $1/(n-1) \leq \kappa$ for $n \geq n_0$. These choices verify every condition in [\(27\)](#), as well as $P_0 + E = X \geq 0$. Although E need not be rational, [Theorem 3.1](#) applies to it in real arithmetic. The analysis in [Appendix D](#) shows that the scaling and the truncated formula can be evaluated with polynomially many bits. Restoring the entropy factor recovers $\text{haf}(A)$. The remaining finitely many dimensions are evaluated exactly. The exact branch returns zero when no perfect matching exists. If one exists, both branches return a positive number. $\square$

The normalization identity and symmetric dilation above reduce the permanent to [Theorem 3.1](#). The bounds on the factors supply the entrywise bound and the singular-value gap supplies the norm bound. The formal reduction is recorded in [Appendix B](#), and its finite-precision implementation follows from [Appendix D](#).

This completes the normalization stage. In either model, the original count is an explicit entropy factor times $\text{haf}(P + E)$, where $P \in \{P_0, P_b\}$ and E satisfies [\(27\)](#) with constants determined only by the fixed input parameters.

3.3. An inverse-Gamma representation. The proof of [Theorem 3.1](#) begins with an exact one-dimensional representation. Fix one of the two reference matrices P from [Section 3.1](#) and a perturbation E satisfying [\(27\)](#). Expanding $\text{haf}(P + E)$ according to the edges chosen from E leaves coefficients that depend only on the number of chosen edges. We identify these coefficients with negative moments of one Gamma random variable, obtaining an average of a signed matching generating polynomial.

Fix $\beta \geq 0$ and $\kappa > 0$, and suppose that E satisfies [\(27\)](#). If $\kappa = 1$, then $E = 0$ and the hafnian is explicit. If $\kappa > 1$, there are no admissible inputs. We therefore assume $0 < \kappa < 1$. To treat the two reference matrices in one formula, let $\tau = 0$ for the complete-graph reference $P = P_0$, and let $\tau = 1$ for the complete-bipartite reference $P = P_b$. Define $a_n = \frac{n+1+\tau}{2}$, $b_n = \frac{n-1+\tau}{2}$, and let U be a Gamma random variable with shape a_n and rate b_n . The variable U therefore has mode $(a_n - 1)/b_n = 1$ in both cases. Thus the Gamma mass is centered at the parameter value where the subsequent subset expansion is controlled. For a matching M on $[n]$, let $\text{wt}_E(M)$ denote the product of the entries E_{uv} over its edges. Define the weighted matching generating polynomial

$$Z_E(t) = \sum_{M \text{ a matching}} t^{|M|} \text{wt}_E(M).$$

We normalize the desired hafnian by its reference value and write $Q_E = \frac{\text{haf}(P+E)}{\text{haf}(P)}$.

Lemma 3.14 (Inverse-Gamma identity). *The normalized hafnian has the exact representation*

$$(36) \quad Q_E = \mathbb{E} Z_E(U^{-1}).$$

For the complete-graph reference, $\text{haf}(P) = (n-1)!!(n-1)^{-n/2}$. For the complete-bipartite reference, where $n = 2m$, one has $\text{haf}(P) = m!/m^m$.

Proof. We compare the coefficients of Z_E with the negative moments of U , treating the two reference matrices separately.

For $0 \leq k \leq n/2$, let $m_k(E)$ be the sum of $\text{wt}_E(M)$ over all k -matchings. In the complete-graph case, expanding according to the edges on which E is selected gives

$$\begin{aligned} \text{haf}(P_0 + E) &= \sum_{k=0}^{n/2} (n-1)^{-(n/2-k)} (n-2k-1)!! m_k(E), \\ \mathbb{E} U^{-k} &= \left(\frac{n-1}{2} \right)^k \frac{\Gamma((n+1)/2 - k)}{\Gamma((n+1)/2)} = (n-1)^k \frac{(n-2k-1)!!}{(n-1)!!}. \end{aligned}$$

Here and below, $(-1)!! = 1$. This proves the identity for P_0 .

Now let $n = 2m$ and $P = P_b$. The block condition on E makes every matching selected from E bipartite. A k -matching leaves $m-k$ vertices on each side, which the reference matrix matches in $(m-k)!$ ways. Consequently,

$$\begin{aligned} \text{haf}(P_b + E) &= \sum_{k=0}^m m^{-(m-k)} (m-k)! m_k(E), \\ \mathbb{E} U^{-k} &= m^k \frac{\Gamma(m+1-k)}{\Gamma(m+1)} = m^k \frac{(m-k)!}{m!}. \end{aligned}$$

Here U has shape $m+1$ and rate m . Since $\text{haf}(P_b) = m!/m^m$, substitution into Z_E proves the bipartite case. $\square$

The inverse-Gamma identity reduces the matrix problem to controlling $Z_E(t)$ for parameters t near one. We represent this polynomial by Wick's rule, using the complex Gaussian convention introduced at the beginning of [Section 2](#). In particular, $A_E = E_+^{1/2} + iE_-^{1/2}$ has bilinear second-moment matrix E .

Let $\xi = A_E x$ and define the energy $\mathcal{E}_E(x) = x^\top |E| x$. Wick's rule gives

$$Z_E(t) = \mathbb{E}_x \prod_{i=1}^n (1 + \sqrt{t} \xi_i) \quad (t \geq 0).$$

Since $E\mathbf{1} = 0$, one has $A_E\mathbf{1} = 0$ and hence $\sum_i \xi_i = 0$. The squared moduli of the factors therefore satisfy $\sum_{i=1}^n |1 + \sqrt{t} \xi_i|^2 = n + t \sum_{i=1}^n |\xi_i|^2 = n + t x^\top |E| x$. Applying AM-GM now gives

$$(37) \quad |Z_E(t)| \leq \mathbb{E}_x \left(1 + \frac{t \mathcal{E}_E(x)}{n} \right)^{n/2}.$$

The density of U is

$$(38) \quad p_{n,\tau}(u) = \frac{b_n^{a_n}}{\Gamma(a_n)} u^{a_n-1} e^{-b_n u} \quad (u > 0).$$

For an event $\mathcal{A}$, the notation $\mathbf{1}_{\{\mathcal{A}\}}$ denotes its indicator.

Proposition 3.15 (Localization of the Gamma average). *Fix $0 \leq \beta < \infty$ and $0 < \kappa < 1$. Suppose that E satisfies the four conditions in [\(27\)](#). Choose fixed numbers r_0 and r_1 with $0 < 1 - \kappa < r_0 < 1 < r_1$. There are constants $c > 0$ and n_0 , depending only on β, κ, r_0, r_1 , such that*

$$(39) \quad \mathbb{E} [|Z_E(U^{-1})| \mathbf{1}_{\{U \notin [r_0, r_1]\}}] \leq e^{-cn}$$

whenever $n \geq n_0$.

The upper and lower tails are controlled by combining (37) with the corresponding Gamma Chernoff bounds. The Gaussian factor contributes only $e^{O(\sqrt{n})}$, while the Gamma density has a strictly positive linear rate away from its mode at one. The details are given in Section C.4.

Henceforth we work only on the compact parameter interval $\mathcal{I} = [1/r_1, 1/r_0]$. For every $t \in \mathcal{I}$, $t\|E\| \leq \frac{1-\kappa}{r_0} < 1$. This strict margin permits the Gaussian change of measure used next.

3.4. Matching-specific Gaussian completion. The goal of this subsection is to turn every retained parameter value into an admissible Gaussian product, uniformly over the localization interval, and then derive a real formula for each retained coefficient.

The Wick representation still contains one factor $1 + \sqrt{t}\xi_i$ at every vertex. Centering removes their combined linear term. We then absorb the quadratic term into the Gaussian measure, leaving a residual factor whose Taylor expansion begins in degree three. For $t \in \mathcal{I}$, define the transformed matrix, the corresponding single-coordinate factor, and its nonconstant part by

$$K_t = tE(I + tE)^{-1}, \quad g(z) = (1 + z)e^{-z+z^2/2}, \quad f(z) = g(z) - 1.$$

The key point is that $f(z) = O(z^3)$ at the origin.

Lemma 3.16 (Matching scalar factor). *The function g is admissible with envelope weights $a_R = 1$ and $a_I = 0$, vanishing order $\ell = 3$, $C_f = 3$, and $\delta_f = 1/2$.*

The elementary proof is given in Appendix C.1.

Lemma 3.17 (Gaussian change of measure). *For every $t \in \mathcal{I}$,*

$$(40) \quad Z_E(t) = \det(I + tE)^{-1/2} \mathbb{E}_{K_t} \prod_{i=1}^n g(\phi_i).$$

The square root is the positive one.

Proof. Put $C = tE$ and let $\psi = (C_+^{1/2} + iC_-^{1/2})x$. Wick's rule gives $Z_E(t) = \mathbb{E}_x \prod_{i=1}^n (1 + \psi_i)$. Since $E\mathbf{1} = 0$, spectral calculus gives $C_+^{1/2}\mathbf{1} = C_-^{1/2}\mathbf{1} = 0$. Hence $\sum_i \psi_i = 0$. The positive and negative spectral parts have disjoint supports, so $\sum_i \psi_i^2 = x^\top Cx$. The definition of g now gives

$$\prod_{i=1}^n (1 + \psi_i) = e^{-\frac{1}{2}x^\top Cx} \prod_{i=1}^n g(\psi_i).$$

Since $\|C\| < 1$, the matrix $I + C$ is positive definite. Absorbing the quadratic factor into the Gaussian density and using $y = (I + C)^{1/2}x$ contributes $\det(I + C)^{-1/2}$. With $K = C(I + C)^{-1}$, spectral calculus gives $A_C(I + C)^{-1/2} = A_K$, proving (40).

Every positive eigenvalue of K is smaller than $1/2$. Therefore Lemma 3.16 gives

$$\mathbb{E}_y \prod_i |g((A_K y)_i)| \leq \mathbb{E}_y e^{y^\top K y} = \det(I - 2K_+)^{-1/2} < \infty.$$

This justifies the change of variables absolutely. The square root is the positive one because $I + C$ is positive definite. $\square$

The expansion over coordinate subsets requires three facts uniformly in t : the transformed complex Gaussian integral retains an integrability margin, the entries of K_t remain $O(1/n)$, and the determinant prefactor stays of constant order. Put $\rho_0 = 1 - \kappa$ and $q_0 = \frac{\rho_0}{r_0} < 1$. Thus q_0 is a uniform upper bound for $t\|E\|$ on $\mathcal{I}$.

Lemma 3.18 (Uniform bounds for K_t). *Under the hypotheses and notation of Proposition 3.15, there are constants $0 < d \leq 1$, $\beta' < \infty$, and $C < \infty$ that depend only on β, κ, r_0, r_1 , such that*

$$(41) \quad \begin{aligned} I - 2K_t &\succeq dI, \quad \max_{i,j} |(K_t)_{ij}| \leq \frac{\beta'}{n}, \\ e^{-C} &\leq \det(I + tE)^{-1/2} \leq e^C. \end{aligned}$$

for every $t \in \mathcal{I}$.

Proof. The spectral calculus, resolvent, and trace estimates are recorded in Section C.6. $\square$

Lemma 3.16 shows that g is admissible with envelope weights $a_R = 1$ and $a_I = 0$, vanishing order $\ell = 3$, and the fixed local constants $C_f = 3$ and $\delta_f = 1/2$. By Lemma 3.18, the matrices K_t satisfy the diffuse entrywise bound and $I - 2K_t \succeq dI$. Since $d \leq 1$, spectral calculus gives $I - 2(K_t)_+ \succeq dI$. Thus the matrices K_t are admissible with fixed β' and d , uniformly for $t \in \mathcal{I}$. The truncation theorem in Theorem 2.3 therefore applies throughout this interval. We retain its notation $W_{K_t}(S)$ for the contribution of a coordinate set S .

3.4.1. Evaluation of a retained subset. The truncation theorem identifies the subsets that the algorithm retains. We now evaluate the contribution of one such subset without enumerating Gaussian pairings. Because $f = g - 1$, inclusion–exclusion reduces $W_K(S)$ to Gaussian products of g , and a weighted monomer–dimer recurrence evaluates each product.

Proposition 3.19 (Fixed-subset evaluation). *Let K be a real symmetric matrix. Suppose that S has size s and $\|K[S]\|_{\text{row}} \leq 1/4$. Then $W_K(S)$ can be computed in $3^s \text{poly}(s)$ arithmetic and elementary-function operations.*

Proof. We first evaluate the product of the residual factors on a fixed subset T , using a Gaussian shift and a monomer–dimer recurrence. Inclusion–exclusion then recovers $W_K(S)$.

For $T \subseteq S$, write $G_K(T) = \mathbb{E}_{K[T]} \prod_{i \in T} g(\phi_i)$, $G_K(\emptyset) = 1$. For nonempty T , let $C_T = K[T]$ be the restricted bilinear second-moment matrix, let $A_T = A_{C_T}$ be the complex square-root factor defined in Section 2, and put $D_T = I - C_T$, $a_T = A_T^\top \mathbf{1}$. The matrix D_T is the Gaussian precision matrix left after the quadratic term in $\prod_{i \in T} g(\phi_i)$ is absorbed. For every $T \subseteq S$, $\|K[T]\|_{\text{row}} \leq \|K[S]\|_{\text{row}} \leq 1/4$. Hence $\|C_T\| \leq 1/4$ and $D_T \succeq 3I/4$. The definition of g now gives the absolutely convergent integral

$$G_K(T) = (2\pi)^{-|T|/2} \int_{\mathbb{R}^T} e^{-x^\top D_T x / 2 - a_T^\top x} \prod_{i \in T} (1 + (A_T x)_i) dx.$$

Complete the square and set $y = x + D_T^{-1} a_T$. The new contour is the complex translate $\mathbb{R}^T + D_T^{-1} a_T$. The integrand is entire, and the positive definiteness of D_T makes the Gaussian dominate the polynomial on every intermediate contour, so Cauchy’s theorem shifts the coordinates back to $\mathbb{R}^T$. Put $\Sigma_T = D_T^{-1} C_T$ and $\mu_T = -\Sigma_T \mathbf{1}$. Because A_T and D_T are commuting spectral functions of C_T , the shifted affine Gaussian has mean μ_T , bilinear second-moment matrix Σ_T , and normalization exponent $a_T^\top D_T^{-1} a_T = \mathbf{1}^\top \Sigma_T \mathbf{1}$. Define P_T to be the weighted monomer–dimer partition function on T , with monomer weight $1 + (\mu_T)_i$ at i and dimer weight $(\Sigma_T)_{ij}$ on $\{i, j\}$ [HL72]. Explicitly, $P_T(\emptyset) = 1$. For nonempty $U \subseteq T$, choose a fixed $i \in U$ and use

$$P_T(U) = (1 + (\mu_T)_i) P_T(U \setminus \{i\}) + \sum_{j \in U \setminus \{i\}} (\Sigma_T)_{ij} P_T(U \setminus \{i, j\}).$$

Wick’s formula therefore gives

$$(42) \quad G_K(T) = \det(D_T)^{-1/2} \exp\left(\frac{1}{2} \mathbf{1}^\top \Sigma_T \mathbf{1}\right) P_T(T).$$

The row-norm bound gives $\max_{i,j \in S} |K_{ij}| \leq 1/4$ and $I - 2K[S]_+ \succeq I/2$. Thus, as an s by s moment matrix, $K[S]$ satisfies the hypotheses of [Lemma 2.5](#) with $\beta = s/4$ and $d = 1/2$. Consequently, inclusion–exclusion and [Lemma 2.5](#) give

$$W_K(S) = \sum_{T \subseteq S} (-1)^{|S|-|T|} G_K(T).$$

The recurrence for a fixed T uses $2^{|T|} \text{poly}(s)$ operations, and $\sum_{T \subseteq S} 2^{|T|} = 3^s$. $\square$

Although the derivation uses complex Gaussian contours, the evaluation formula itself is real. Indeed, D_T is real positive definite, Σ_T and μ_T are real, and the monomer–dimer recurrence has real weights. Thus the algorithm uses only real arithmetic. The complex square roots and contour shifts are analytic devices used to establish the formula.

3.5. The approximation algorithm. The analytic estimates are now in place. Localization restricts the Gamma average to a fixed interval, and Gaussian completion followed by [Theorem 2.3](#) controls the resulting subset expansion. We now choose the truncation order and approximate the remaining one-dimensional integral.

3.5.1. Truncation, positivity, and evaluation bounds. Apply [Theorem 2.3](#) to K_t using [Lemma 3.18](#), and write c_* for the resulting value of c_{tr} . Uniformly for $t \in \mathcal{I}$, the theorem gives $\|K_t[T]\|_{\text{row}} \leq 1/8$ whenever $|T| \leq c_* n$. The unused margin between $1/8$ and $1/4$ will accommodate finite-precision errors. All constants in this section may depend on the fixed parameters in [Theorem 3.1](#).

For an integer $R \geq 2$ and a parameter $t \in \mathcal{I}$, define the truncated approximation

$$Z_{E,R}(t) = \det(I + tE)^{-1/2} \sum_{|S| < R} W_{K_t}(S).$$

By [\(6\)](#) and [Lemma 3.18](#), there are constants $C, c > 0$ and n_0 such that

$$|Z_E(t) - Z_{E,R}(t)| \leq \left(C \sqrt{\frac{R}{n}} \right)^R + e^{-cn}$$

whenever $n \geq n_0$, $t \in \mathcal{I}$, and $2 \leq R \leq c_* n$. Here we also used the uniform upper bound on the determinant factor from [Lemma 3.18](#).

Because the expansion over coordinate subsets is signed, the additive estimate must be paired with a uniform positive lower bound. Indeed, [\(7\)](#), applied uniformly to K_t , gives $\mathcal{Z}_{K_t} = 1 + O(n^{-1})$. Together with the determinant bounds in [Lemma 3.18](#), this gives constants $0 < a_{\text{pos}} \leq b_{\text{pos}} < \infty$ such that

$$(43) \quad a_{\text{pos}} \leq Z_E(t) \leq b_{\text{pos}} \quad (t \in \mathcal{I}), \quad a_{\text{pos}} \leq Q_E \leq b_{\text{pos}}$$

for all sufficiently large n , after adjusting the two constants. To obtain the bounds for Q_E , average the first pair of inequalities over $U \in [r_0, r_1]$ and use [Proposition 3.15](#) for the discarded Gamma tails.

Corollary 3.20 (Perfect-matching existence). *For every fixed choice of the parameters in [Theorem 3.3](#), there is an n_0 such that the support of every input covered by that theorem with $n \geq n_0$ contains a perfect matching.*

Proof. The entropy-scaled matrix is $X = P_0 + E$ and has the same support as the input. By [\(43\)](#), $\text{haf}(X) = \text{haf}(P_0)Q_E > 0$ once $n \geq n_0$. Thus the support contains a perfect matching. In particular, only the exact branch of the general criterion can encounter a zero hafnian. $\square$

We next collect the quantitative estimates used by the algorithm. Recall the Gamma density $p_{n,\tau}$ from (38). For an integer $R \geq 2$, define

$$\varphi_R(u) = p_{n,\tau}(u)Z_{E,R}(u^{-1}), \quad Q_R = \int_{r_0}^{r_1} \varphi_R(u) du.$$

Thus φ_R is the truncated Gamma-average integrand, and Q_R is its integral over the retained interval.

There are constants $C, c > 0$, $0 < \eta_0 \leq c_*/2$, and n_0 such that, whenever $n \geq n_0$ and $2 \leq R \leq 2\eta_0 n$,

$$(44) \quad |Q_E - Q_R| \leq \left(C \sqrt{\frac{R}{n}} \right)^R + e^{-cn}.$$

Moreover,

$$C^{-1} \leq Q_E \leq C, \quad \|\varphi'_R\|_{L^\infty([r_0, r_1])} \leq n^C \left(\frac{3n}{R} \right)^{CR}.$$

At a specified point $u \in [r_0, r_1]$, the expression $\varphi_R(u)$ has a real-arithmetic evaluation circuit of size at most $n^C (3n/R)^{CR}$. The truncation estimate follows from localization and the uniform estimate above. For the remaining bounds, the resolvent formulas show that K_t and K'_t have entries $O(n^{-1})$. If $s = |T| < R$, then $\|K_t[T]\|_{\text{row}} \leq 1/4$ and $\|(I - K_t[T])^{-1}\|_{\text{row}} \leq 4/3$, and the derivatives of the shifted covariance and mean are $O(1)$. Differentiating the fixed-subset formula and its recurrence therefore bounds a fixed T contribution and its derivative by $e^{O(s)} \text{poly}(s)$, with $2^s \text{poly}(s)$ operations. Inclusion-exclusion costs $3^s \text{poly}(s)$, and summing over $|S| < R$ gives the claimed bound. Finally, $\|p_{n,\tau}\|_\infty = O(\sqrt{n})$ and $\|p'_{n,\tau}\|_\infty = O(n^{3/2})$ by Stirling's formula, so the same bound holds for φ'_R .

3.5.2. Choosing the cutoff and quadrature. We now choose the subset cutoff so that the analytic tail is smaller than the requested error, and then choose the quadrature mesh from the derivative bound.

We may assume that $C \geq 12$. Put

$$b = \left\lceil \log_2 \frac{C^2}{\varepsilon} \right\rceil,$$

and choose $\eta > 0$ no larger than η_0 , $c/(4 \log 2)$, and the constant supplied by Lemma 2.10 for $\alpha = 1/2$, $C_2 = C$, and $c_{\text{tr}} = 2\eta_0$. We may decrease η further without changing that lemma. If $n < n_0$ or $b \geq \eta n$, we use exact evaluation. Otherwise, let $R \geq 2$ be the least integer satisfying

$$(45) \quad \left(\frac{C^2 R}{n} \right)^R \leq 2^{-2b}.$$

The algorithm finds this least integer by testing successive values of R . The cutoff condition is equivalently $(C\sqrt{R/n})^R \leq 2^{-b}$. Hence Lemma 2.10 shows that R exists, satisfies $R \leq 2\eta_0 n$, and obeys

$$(46) \quad R \log \frac{en}{R} = O(b + \log n).$$

The cutoff and the choice of η also give $(C\sqrt{R/n})^R \leq 2^{-b}$ and $e^{-cn} \leq 2^{-4b}$, and the derivative and circuit bounds in the preceding display are $n^{O(1)} 2^{O(b)}$.

For the quadrature, divide $[r_0, r_1]$ into L equal subintervals and use their midpoints $u_1, \dots, u_L$. Choose L to be a sufficiently large constant multiple of $2^b n^C (3n/R)^{CR}$, rounded up to an integer. With $h = (r_1 - r_0)/L$, put

$$(47) \quad Q_{R,L} = h \sum_{\ell=1}^L \varphi_R(u_\ell).$$

The elementary Lipschitz midpoint estimate and the derivative estimate above give $|Q_{R,L} - Q_R| \leq 2^{-b}$. Thus the truncation, Gamma tail, and quadrature errors together are at most $3 \cdot 2^{-b}$. The number of nodes and the work at each node are polynomial in n and $1/\varepsilon$ by (46).

3.5.3. *The exact branch.* For a symmetric weighted matrix Y and a vertex set S , let $F_Y(S)$ be the hafnian of the principal matrix $Y[S]$, with $F_Y(\emptyset) = 1$. For nonempty even S , choose its least vertex v and use the standard subset recurrence

$$F_Y(S) = \sum_{u \in S \setminus \{v\}} Y_{uv} F_Y(S \setminus \{u, v\}).$$

The recurrence has $2^{O(n)}$ states. If $b \geq \eta n$, then $2^{O(n)} \leq 2^{O_{\beta, \kappa}(b)} = \varepsilon^{-O_{\beta, \kappa}(1)}$. The other exact branch has $n < n_0$, a fixed threshold depending only on β and κ . Thus both branches have polynomial real-arithmetic complexity in n and $1/\varepsilon$.

For rational inputs the recurrence is applied to the original matrix rather than to the generally irrational entropy scaling. Its bit implementation, and the zero-output case, are recorded in [Appendix D](#).

Proof of Theorem 3.1. We verify accuracy and arithmetic cost in the approximation branch, then invoke the subset recurrence for the complementary parameter range. In the approximation branch, evaluate (47). At each node, compute the weights $W_{K_t}(S)$ for $|S| < R$ by [Proposition 3.19](#). The number of arithmetic operations is bounded by

$$L \sum_{s < R} \binom{n}{s} 3^s \text{poly}(n) \leq L \exp \left(O \left(R \log \frac{en}{R} \right) \right) \text{poly}(n).$$

By (46) and the choice of L , this is $n^{O(1)} 2^{O(b)} = n^{O(1)} \varepsilon^{-O(1)}$. The estimates of the preceding subsection show that the resulting value differs from Q_E by at most $3 \cdot 2^{-b} \leq 3\varepsilon/C^2$. Since $Q_E \geq C^{-1}$, this is a relative error of at most $3\varepsilon/C \leq \varepsilon/4$, which lies within the requested factor $e^{\pm\varepsilon}$. Multiplication by the explicit value of $\text{haf}(P)$ gives the requested hafnian. The exact branch has already been discussed, so the theorem follows. $\square$

The preceding proof is in real arithmetic. The entropy scaling and the truncated formula are stable under polynomially small perturbations, and the matrices inverted by the evaluators remain uniformly well conditioned. Consequently, polynomially many working bits suffice. The complete bit-model argument is given in [Appendix D](#).

We finish the matching section by deriving the main theorem and the three corollaries stated in the introduction.

Proof of Theorem 1.1(a). All bounded exceptional dimensions arising in the reduction are handled by exact evaluation. Above the resulting fixed threshold, [Lemma 3.8](#) shows that every edge of the support graph belongs to a perfect matching. The factor bounds in (30) hold with $c_{\text{lo}} = \theta\sqrt{2\gamma}$ and $c_{\text{hi}} = (\theta\sqrt{2\gamma})^{-1}$. Moreover, [Lemma 3.11](#) gives $\lambda_2(L_G) \geq 2\gamma n$ and $Q_G \geq 2\gamma n I$. Thus [Theorem 3.3](#), with $q' = q = 2\gamma$, gives the stated algorithm and bit complexity. $\square$

Proof of Theorem 1.1(b). The finitely many dimensions below the threshold in [Proposition 3.13](#) are handled exactly. In the remaining dimensions, that proposition gives total support, the factor bounds (34), and $\sigma_2(Y^*) \leq 1 - 4\gamma^3\theta^6$. Therefore [Theorem 3.4](#), with the fixed factor bounds from (34) and $\kappa_b = 4\gamma^3\theta^6$, gives the stated algorithm and bit complexity. $\square$

Proof of Corollary 1.3. Apply [Theorem 1.1](#) with, for example, $\gamma = 1/4$. Complete supports satisfy the required degree inequalities in all sufficiently large dimensions, and the remaining dimensions are handled exactly. $\square$

Proof of Corollary 1.4. Let $n = |V(G)|$ and take $\gamma' = \gamma/2$. For every vertex v , the graph $G - v$ satisfies $\delta(G - v) \geq (1/2 + \gamma)n - 1 \geq (1/2 + \gamma')(n - 1)$ when n is sufficiently large in terms of γ . Apply [Theorem 1.1\(a\)](#) to every $G - v$ and add the estimates. Each near-perfect matching occurs once, according to its unmatched vertex. A sum of positive $e^{\pm\varepsilon}$ -approximations has the same relative guarantee. The remaining bounded values of n are handled exactly. $\square$

Proof of Corollary 1.5. The entropy normalization and the definition of Q_E give

$$\text{haf}(A) = e^{h_A(G)} \text{haf}(P_0) Q_E, \quad \text{haf}(P_0) = (n-1)!!(n-1)^{-n/2}.$$

(43) bounds Q_E above and below by positive constants that depend only on γ and θ . Decreasing the lower constant $a_{\gamma,\theta}$ and increasing the upper constant $b_{\gamma,\theta}$ handles the finitely many values of n sent to the exact branch. Every admissible support has a perfect matching by Dirac's theorem. There are only finitely many such supports in these dimensions, and compactness of $[\theta, 1]^{E(G)}$ gives uniformity over the weights for each support. This proves the first assertion. Stirling's formula gives $\log \text{haf}(P_0) = -n/2 + O(1)$. Taking logarithms proves the equivalent form. $\square$

4. THE ZERO-FIELD ISING MODEL WITH DIFFUSE COUPLINGS

The Ising application is shorter because no entropy scaling or Gamma average is needed. Hubbard–Stratonovich converts the spin sum into a product of hyperbolic cosines. Quadratic completion then produces the same Gaussian-product form as in the matching argument, but with growth in the imaginary rather than the real direction. We first verify the required one-variable bound. We then prove the representation, bound the completed moment matrix, and evaluate the retained subsets.

Let J be a real symmetric n by n matrix with zero diagonal. We use the normalization

$$(48) \quad Z(J) = \sum_{\sigma \in \{\pm 1\}^n} \exp\left(\frac{1}{2} \sigma^\top J \sigma\right).$$

The diagonal restriction is harmless. An arbitrary diagonal contributes the known factor $\exp(\text{tr } J/2)$.

The one-variable functions needed below are $g_{\text{Is}}(z) = e^{-z^2/2} \cosh z$ and $f_{\text{Is}}(z) = g_{\text{Is}}(z) - 1$.

Lemma 4.1 (Ising scalar factor). *The function g_{Is} satisfies*

$$(49) \quad |g_{\text{Is}}(x + iy)| \leq e^{y^2/2},$$

$$(50) \quad f_{\text{Is}}(z) = -\frac{z^4}{12} + O(z^6) \quad (z \rightarrow 0).$$

It is therefore admissible with envelope weights $a_R = 0$ and $a_I = 1/2$, vanishing order $\ell = 4$, and local constants $C_f = 1/8$ and $\delta_f = 1/2$.

The elementary proof is given in Appendix C.2.

4.1. Gaussian completion. We now remove the spin sum and simultaneously resum the quadratic part of the resulting Gaussian product.

Lemma 4.2 (Hubbard–Stratonovich representation and completion). *Suppose that $\lambda_{\max}(J) < 1$ and define*

$$(51) \quad K = J(I - J)^{-1}.$$

Then

$$(52) \quad Z(J) = 2^n \det(I - J)^{-1/2} \mathbb{E}_K \prod_{i=1}^n g_{\text{Is}}(\phi_i).$$

The square root is the positive one, and the expectation in (52) converges absolutely.

Proof. If x is a standard real Gaussian vector, then, for every complex vector v , one has $\mathbb{E} e^{v^\top x} = e^{v^\top v/2}$. Take $v = A_J \sigma$. Since A_J is symmetric and $A_J^2 = J$, we have $v^\top v = \sigma^\top J \sigma$. Summing the resulting identity over the spin configurations gives $Z(J) = 2^n \mathbb{E}_x \prod_{i=1}^n \cosh((A_J x)_i)$. If $\psi = A_J x$, then $\sum_i \psi_i^2 = x^\top J x$. The definition of g_{Is} therefore gives

$$\mathbb{E}_x \prod_i \cosh(\psi_i) = \mathbb{E}_x e^{x^\top J x/2} \prod_i g_{\text{Is}}(\psi_i).$$

The assumption on J makes $I - J$ positive definite. The change of variables $y = (I - J)^{1/2}x$ absorbs the quadratic exponential into the Gaussian density and contributes $\det(I - J)^{-1/2}$. Spectral calculus gives $A_J(I - J)^{-1/2} = A_K$, where $K = J(I - J)^{-1}$, which proves the identity.

It remains only to justify absolute convergence. Before the change of variables, $\prod_i |\cosh((A_J x)_i)| \leq \exp(x^\top J_+ x/2)$, and $\lambda_{\max}(J_+) < 1$. After the change of variables, (49) gives the majorant $\exp(y^\top K_- y/2)$. If $-a < 0$ is an eigenvalue of J , the corresponding eigenvalue of K_- is $a/(1+a) < 1$. The second Gaussian integral is therefore absolutely convergent as well. $\square$

4.2. The transformed moment matrix. The completed moment matrix retains the diffuse entrywise scale and also satisfies the spectral margin needed below.

Lemma 4.3 (Bounds for the completed moment matrix). *Fix $\beta > 0$ and $0 < \kappa \leq 1$. Suppose that J is real symmetric, has zero diagonal, and satisfies*

$$(53) \quad \max_{i,j} |J_{ij}| \leq \frac{\beta}{n}, \quad \lambda_{\max}(J) \leq 1 - \kappa.$$

For the matrix K in (51), put $\beta_K = \beta + \beta^2/\kappa$ and $d_K = (1 + \beta)^{-1}$. Then

$$(54) \quad \max_{i,j} |K_{ij}| \leq \frac{\beta_K}{n}, \quad I - K_- \succeq d_K I.$$

The negative eigenvalues of K lie in $[-\beta/(1 + \beta), 0]$. Finally,

$$(55) \quad |\log \det(I - J)| \leq \frac{\beta^2}{2\kappa}.$$

Proof. The entrywise hypothesis gives $\|J\| \leq \|J\|_{\text{row}} \leq \beta$. Using $K = J + J(I - J)^{-1}J$ and $\|(I - J)^{-1}\| \leq \kappa^{-1}$, we obtain

$$|(J(I - J)^{-1}J)_{ij}| \leq \|J e_i\|_2 \|(I - J)^{-1}\| \|J e_j\|_2 \leq \frac{\beta^2}{\kappa n}.$$

Here e_i and e_j are standard coordinate vectors. This proves the entrywise part of (54).

The map from the spectrum of J to that of K is $\lambda \mapsto \lambda/(1 - \lambda)$. If $\lambda = -a \leq 0$, then the corresponding eigenvalue of K is $-a/(1 + a)$. Since $a \leq \beta$, this proves the remaining spectral assertions and $I - K_- \succeq d_K I$. In view of (49), these are precisely the entrywise and Gaussian integrability margins required by Theorem 2.3.

For the determinant estimate, let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of J . The zero diagonal gives $\sum_i \lambda_i = 0$, and hence $\log \det(I - J) = \sum_i (\log(1 - \lambda_i) + \lambda_i)$. For $0 \leq \lambda \leq 1 - \kappa$, integration of $t/(1 - t)$ gives $0 \leq -\log(1 - \lambda) - \lambda \leq \lambda^2/(2\kappa)$. For $\lambda = -a \leq 0$, the inequality $0 \leq a - \log(1 + a) \leq a^2/2$ gives the same bound. Thus

$$|\log \det(I - J)| \leq \frac{\|J\|_{\text{F}}^2}{2\kappa} \leq \frac{\beta^2}{2\kappa},$$

as claimed. $\square$

Combining (49) and (50) with Lemma 4.3 shows that the pair (g_{Is}, K) satisfies the hypotheses of Theorem 2.3, with constants depending only on β and κ . Indeed, its envelope matrix is $K_{\text{env}} = K_-/2$, so the required spectral margin is precisely $I - K_- \succeq d_K I$. In particular, writing $\mathcal{Z}_K = \mathbb{E}_K \prod_{i=1}^n g_{\text{Is}}(\phi_i)$, we have

$$(56) \quad \mathcal{Z}_K = 1 + O_{\beta, \kappa}(1/n).$$

4.3. Evaluation of retained subsets. For a subset $T \subseteq [n]$, define $G_K(T) = \mathbb{E}_{K[T]} \prod_{i \in T} g_{\text{Is}}(\phi_i)$. The retained contribution has an explicit sum over the spin configurations on T .

Lemma 4.4 (Retained-subset evaluation). *Let K_T be a real symmetric t by t matrix with $\|K_T\|_{\text{row}} \leq 1/4$. Then*

$$(57) \quad \mathbb{E}_{K_T} \prod_{i=1}^t g_{\text{Is}}(\phi_i) = \det(I + K_T)^{-1/2} 2^{-t} \sum_{\sigma \in \{\pm 1\}^t} \exp\left(\frac{1}{2} \sigma^\top K_T (I + K_T)^{-1} \sigma\right).$$

In particular, the right-hand side is a positive real number, so the full-product terms $G_K(T)$ used below are positive. It is computable using $2^t \text{poly}(t)$ arithmetic operations. Every exponent in (57) has absolute value at most $t/6$.

Proof. Let $\phi = A_{K_T} x$. Expanding the hyperbolic cosines gives

$$\prod_i g_{\text{Is}}(\phi_i) = 2^{-t} \sum_{\sigma \in \{\pm 1\}^t} \exp\left(-\frac{1}{2} x^\top K_T x + \sigma^\top A_{K_T} x\right).$$

Since $I + K_T$ is positive definite, Gaussian integration with a complex linear term gives (57). Here we used $A_{K_T}(I + K_T)^{-1} A_{K_T} = K_T(I + K_T)^{-1}$, which follows from spectral calculus. All matrices in the expression on the right of (57) are real, so the value is positive and real.

The Neumann series gives $\|(I + K_T)^{-1}\|_{\text{row}} \leq 4/3$, and hence $\|K_T(I + K_T)^{-1}\|_{\text{row}} \leq 1/3$. Therefore

$$\left| \frac{1}{2} \sigma^\top K_T (I + K_T)^{-1} \sigma \right| \leq \frac{t}{2} \|K_T(I + K_T)^{-1}\|_{\text{row}} \leq \frac{t}{6}.$$

□

The subset coefficients in the truncation theorem are $W_K(S) = \mathbb{E}_{K[S]} \prod_{i \in S} f_{\text{Is}}(\phi_i)$. By inclusion-exclusion and Lemma 2.5,

$$(58) \quad W_K(S) = \sum_{T \subseteq S} (-1)^{|S| - |T|} G_K(T).$$

Thus a direct evaluation of $W_K(S)$ takes $3^{|S|} \text{poly}(|S|)$ operations: after a set T is chosen, (57) sums over its $2^{|T|}$ spin configurations, and $\sum_{T \subseteq S} 2^{|T|} = 3^{|S|}$.

4.4. The approximation scheme. We can now prove the main result.

Proof of Theorem 1.2. The asymptotic statements follow at once from (52), (55) and (56). We prove the algorithmic assertion.

Apply Theorem 2.3 with the fixed Ising factor, the entrywise constant β_K , and the margin d_K . Denote its constants by $C_1, c_{\text{tr}}, a_{\text{tail}} > 0$ and n_0 . Increase n_0 , if necessary, so that

$$(59) \quad \mathcal{Z}_K \geq \frac{1}{2} \quad (n \geq n_0).$$

Set $C_{\text{cut}} = 2C_1$ and $b = \lceil \log_2(16/\varepsilon) \rceil$. Fix a sufficiently small constant $\eta > 0$, depending only on β and κ , for which Lemma 2.10 applies with $\alpha = 1$ and $C_2 = C_{\text{cut}}$, and such that $\eta \leq a_{\text{tail}}/(3 \log 2)$.

We first dispose of the enumeration branch. If $n < n_0$ or $b \geq \eta n$, evaluate the positive sum in (48) directly. In the second case, $2^n \leq 2^{b/\eta}$, while in the first case n is bounded by a constant. Moreover, $|\sigma^\top J \sigma|/2 \leq \beta n/2$. Standard evaluation of the positive exponential terms to a common relative error below $\varepsilon/4$ therefore gives the required estimate in $2^{O_{\beta, \kappa}(b)} \text{poly}(L_J + b)$ bit operations.

Suppose from now on that $n \geq n_0$ and $b < \eta n$. Let $R \geq 2$ be the least integer satisfying

$$(60) \quad \left(C_{\text{cut}} \frac{R}{n}\right)^R \leq 2^{-b}.$$

By Lemma 2.10, after decreasing η once, this integer exists and satisfies $R \leq c_{\text{tr}}n$. Minimality gives

$$(61) \quad R \log \frac{en}{R} = O_{\beta, \kappa}(b + \log n).$$

Since $R \geq 2$, replacing C_1 by $C_{\text{cut}} = 2C_1$ divides the first term of the truncation bound by at least four. Our choice of η gives $e^{-a_{\text{tail}}n} \leq 2^{-3b}$. Hence

$$\begin{aligned} \left| \mathcal{Z}_K - \sum_{|S| < R} W_K(S) \right| &\leq \left(C_1 \frac{R}{n} \right)^R + e^{-a_{\text{tail}}n} \\ &\leq 2^{-b-2} + 2^{-3b} \leq 2^{-b-1}. \end{aligned}$$

Compute the retained sum by (57) and (58) to absolute error at most 2^{-b-2} , and call the result $\tilde{\mathcal{Z}}_K$. Then

$$(62) \quad |\tilde{\mathcal{Z}}_K - \mathcal{Z}_K| \leq 3 \cdot 2^{-b-2}.$$

Approximate the positive prefactor $2^n \det(I - J)^{-1/2}$ to relative error at most 2^{-b-2} and multiply it by $\tilde{\mathcal{Z}}_K$. From (59) and (62), the relative error in $\tilde{\mathcal{Z}}_K$ is at most $3 \cdot 2^{-b-1}$. In particular, $\tilde{\mathcal{Z}}_K > 0$, because $\mathcal{Z}_K \geq 1/2$ and $3 \cdot 2^{-b-2} < 1/2$. The relative error in the final product is therefore less than $2^{1-b} \leq \varepsilon/8$. Since $0 < \varepsilon < 1$, this is contained in the interval $[e^{-\varepsilon}, e^{\varepsilon}]$.

Finally, (61) gives $\sum_{s < R} \binom{n}{s} 3^s = n^{O_{\beta, \kappa}(1)} 2^{O_{\beta, \kappa}(b)}$. Together with Lemma 4.4, this proves the polynomial real-arithmetic bound. The bit implementation is given in Appendix D, completing the proof. $\square$

5. SCOPE AND OPEN PROBLEMS

We close with three limitations of the present method and the questions they suggest.

1. *The truncation principle.* The entrywise $O(1/n)$ hypothesis is intrinsic to the present truncation argument and is not implied by the spectral margin, as simple rank-one examples show. This prevents a direct application to sparse expanders, where a graph-specific resummation or correlation estimate would be needed. Both applications also eliminate a linear mode, through $E\mathbf{1} = 0$ for matchings and zero external field for Ising. We do not know whether a surviving finite-rank mode can be integrated separately while the theorem is applied to the centered fluctuations. Such an extension would require uniform envelope and evaluation bounds that we do not yet have.

2. *Perfect matchings.* The present method does not reach the exact graph Dirac threshold. There the entropy scaling may place order-one weight on an edge. Indeed, split $n = 2m$ vertices into two m -vertex sets U and V , include all cross edges, all edges within V , and one edge within U . If t is the entropy-maximizing weight of that last edge, symmetry gives $t/(1-t) = \sqrt{(m-1)/(2m)}$, so $t \rightarrow \sqrt{2} - 1$. Separately, a copy of $K_{m,m}$ with one added edge in each part shows that the smallest scaled eigenvalue can approach -1 . Whether a deterministic FPTAS exists at the endpoint for all Dirac graphs remains open.

A separate argument gives a deterministic FPTAS for the structured endpoint family obtained by replacing the edge inside U by a graph F of maximum degree at most a fixed constant Δ . Let $m_k(F)$ denote the number of k -edge matchings in F , and let μ be the maximum size of a matching in F . Then $\# \text{PM} = m! \sum_{k=0}^{\mu} m_k(F)/(2^k k!)$. If $\mu \geq \max\{8e^2\Delta, 2\log_2(8/\varepsilon)\}$, the bound $|E(F)| \leq 2\Delta\mu$ makes the terms with $k > \mu/2$ contribute at most $\varepsilon/4$ of the sum, while the algorithm of Jain, Perkins, Sah, and Sawhney [JPSS22], with $\delta = 1/2$, approximates all terms with $k \leq \mu/2$. Otherwise, a dynamic program over the endpoints of a maximum matching evaluates the sum exactly in $2^{O(\mu)} \text{poly}(m) = \text{poly}(m, 1/\varepsilon)$ time.

Below the half-degree threshold, Corollary 3.6 controls unweighted pseudorandom supports, but the analogous weighted statement remains open in the full degree-codegree range. Our counting

algorithm also does not immediately give a sampler, since conditioning on k matching edges reduces the guaranteed degree surplus from γn to $\gamma n - k$.

3. The Ising model. The bounded-remainder conclusion in [Theorem 1.2](#) also requires diffuseness. A single fixed bond of strength $q \in (0, 1)$ gives the normalized factor $\sqrt{1 - q^2} \cosh q$, independently of the number of isolated spins, so the $1 + O(1/n)$ conclusion fails without the entrywise hypothesis. Deterministic approximation under a spectral hypothesis alone remains open. Our argument also does not reach the $n^{-1/2}$ interaction scale of dense spin glasses: the effective entrywise parameter grows like $\sqrt{n}$, and the small-support bound no longer survives the sum over supports.

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APPENDIX A. PROOFS OF THE SCALING-CRITERION CONSEQUENCES

This appendix supplies the deductions deferred from the consequences of the scaling criteria. The first proof treats the regular cases, whose stochastic normalization is explicit. The remaining argument establishes the symmetric degree-codegree consequence.

A.1. Regular inputs.

Proof of Corollary 3.5. For part (a), the maximum-entropy scaling is $X = A_G/d$, with $r_v = d^{-1/2}$ and $n^{-1/2} \leq r_v \leq c^{-1/2}n^{-1/2}$. The spectral hypothesis gives $\lambda_2(L_G) \geq \kappa cn$ and $Q_G \succeq \kappa cnI$. Apply Theorem 3.3 with $\theta = 1$, $c_{lo} = 1$, $c_{hi} = c^{-1/2}$, and $q' = q = \kappa c$.

For part (b), regularity gives total support and the Sinkhorn scaling $Y^* = B/d$, with $r_i = c_j = d^{-1/2}$. The bounds are again between $m^{-1/2}$ and $c^{-1/2}m^{-1/2}$. Apply Theorem 3.4 with $\theta = 1$, $c_{lo} = 1$, $c_{hi} = c^{-1/2}$, and $\kappa_b = \kappa$. $\square$

A.2. Symmetric degree and codegree bounds.

Proof of Corollary 3.6. The edge-extension hypothesis makes the maximum-entropy scaling positive on every edge. Write it as $X_{uv} = r_u r_v$, and put $d_- = (1-\rho)p$, $d_+ = (1+\rho)p$, $R_0 = \frac{d_-}{\eta-2\rho p}$. At vertices a, b carrying the largest and smallest factors, subtraction of the row equations and the degree bounds give $1/r_b - 1/r_a \leq (d_+ - \eta)nr_a$ and $1/(d_+n) \leq r_a r_b \leq 1/(d_-n)$. Multiplying the first inequality by r_b and using the second yields $r_{\max}/r_{\min} \leq R_0$. Consequently, $1/\sqrt{R_0 d_+ n} \leq r_v \leq \sqrt{R_0/(d_- n)}$ for every vertex v , and hence $1/(R_0 d_+ n) \leq X_{uv} \leq R_0/(d_- n)$ on every edge.

For the Laplacian gap, sum $(z_u - z_w)^2 \leq 2(z_u - z_v)^2 + 2(z_v - z_w)^2$ over common neighbors v of each pair u, w . Every squared edge difference is counted at most $2n$ times. Since $\sum_{u < w} (z_u - z_w)^2 = n\|z\|_2^2$ for $z \perp \mathbf{1}$, this gives

$$z^\top L_G z \geq \frac{\eta n}{4} \|z\|_2^2, \quad \lambda_2(L_G) \geq \frac{\eta n}{4}.$$

The displayed factor bounds, this Laplacian estimate, and $Q_G \succeq qnI$ verify the hypotheses of Theorem 3.3, which proves the algorithmic claim.

For $G(n, p)$, take $\rho = p/8$, $\eta = p^2/2$, and any fixed $q < p/2$. Chernoff bounds and the Füredi–Komlós estimate give the degree, codegree, and signless-Laplacian conditions with high probability [FK81]. For a fixed pair u, v , put $s = (n-2)/2$ and partition the remaining vertices into two s -sets. Hall’s condition gives

$$(63) \quad \Pr\{G(s, s, p) \text{ has no perfect matching}\} \leq \sum_{k=1}^s \binom{s}{k} \binom{s}{k-1} (1-p)^{k(s-k+1)} = e^{-\Omega_p(s)}.$$

Pairing k with $s-k+1$ bounds the sum by a geometric series with ratio $s^2(1-p)^{s/2}$, proving the last equality. A union bound over all pairs then shows that every present edge extends to a perfect matching. $\square$

APPENDIX B. BIPARTITE SCALING DETAILS

The permanent reduction uses only standard scaling and spectral estimates. For reference, we give the details here in the order in which they are invoked in the main text.

B.1. Entropy normalization.

Proof of Proposition 3.12. Total support lets us average supported permutation matrices to obtain a feasible point positive on every supported entry. The boundary argument from Proposition 3.9 puts Y^* in the same relative interior, and the row and column multiplier equations give $Y_{ij}^* = b_{ij} r_i c_j$. Moreover, $\sum_i \log r_i + \sum_j \log c_j = \sum_{i,j} Y_{ij}^* \log(Y_{ij}^*/b_{ij}) = -h_B$. Every supported permutation uses each row and column factor once. Its term in $\text{per}(B)$ is therefore e^{h_B} times its term in $\text{per}(Y^*)$. Summing over supported permutations proves (32). $\square$

B.2. Fixed-margin Sinkhorn estimates.

Proof of Proposition 3.13. Put $\alpha = 1/2 + \gamma$, and let G_B denote the bipartite support graph of B . If $b_{ij} > 0$, deleting row i and column j from the support leaves a balanced bipartite graph of minimum degree at least $\alpha m - 1 \geq (m-1)/2$ by the assumed lower bound on m . Hall's theorem therefore extends ij to a perfect matching. Thus B has total support.

For row sets S and column sets T , write $Y^*(S, T) = \sum_{i \in S, j \in T} Y_{ij}^*$, $r(S) = \sum_{i \in S} r_i$, and $c(T) = \sum_{j \in T} c_j$. Let U_r and U_c consist of the $\lfloor (1 - \alpha)m \rfloor$ largest row and column factors, and let W_r and W_c be their complements. Stochasticity and $Y_{ij}^* \leq r_i c_j$ give

$$Y^*(W_r, W_c) = |W_r| - Y^*(W_r, U_c) \geq |W_r| - |U_c| \geq 2\gamma m, \quad r(W_r)c(W_c) \geq 2\gamma m.$$

Write $r_{\max} = \max_i r_i$, $r_{\min} = \min_i r_i$, and define $c_{\max}, c_{\min}$ analogously. The nonneighbors of any row have at most $|U_c|$ elements. The row and column equations, together with the preceding mass bound, give the extremal products as follows. The neighboring column-factor sum is at least $c(W_c)$, and symmetrically the neighboring row-factor sum is at least $r(W_r)$. Thus $r_{\max} \leq 1/(\theta c(W_c))$ and $c_{\max} \leq 1/(\theta r(W_r))$. Conversely, $b_{ij} \leq 1$ gives $r_i \geq 1/(m c_{\max})$ and $c_j \geq 1/(m r_{\max})$. Hence $r_{\max} c_{\max} \leq 1/(2\gamma \theta^2 m)$ and $r_{\min} c_{\min} \geq 2\gamma \theta^2/m$. Since $Y_{ij}^* = b_{ij} r_i c_j$ with $b_{ij} \in [\theta, 1]$ on the support, these estimates prove (33). They also give the looser uniform range $2\gamma \theta^3/m \leq r_i c_j \leq 1/(2\gamma \theta^3 m)$ on every supported entry. Any two rows share a supported column, and any two columns share a supported row. Taking ratios through these common entries shows that the ratio of any two row factors, and likewise of any two column factors, is bounded by a constant depending only on γ and θ .

Put $\ell_0 = \log(2\gamma \theta^3)$ and $u_0 = \log(1/(2\gamma \theta^3))$. On every supported entry, $\ell_0 \leq x_i^* + y_j^* \leq u_0$. The gauge gives equal row and column means, while the ratio bound shows that every coordinate differs from its mean by a constant depending only on γ and θ . Applying these facts at one supported entry places every coordinate in a fixed interval depending only on these two parameters. Exponentiating proves (34).

The symmetric stochastic matrix $W = Y^*(Y^*)^\top$ controls the remaining singular values. Any two distinct rows of B share at least $2\gamma m$ common support columns. Thus (33) gives $W_{ii'} \geq 8\gamma^3 \theta^6/m$ for $i \neq i'$. For $z \perp \mathbf{1}$,

$$z^\top (I - W) z = \sum_{i < i'} W_{ii'} (z_i - z_{i'})^2 \geq 8\gamma^3 \theta^6 \|z\|_2^2.$$

Hence $\sigma_2(Y^*)^2 \leq 1 - 8\gamma^3 \theta^6$, which implies (35). $\square$

B.3. The bipartite scaling criterion.

Proof of Theorem 3.4. By Proposition 3.12, $\text{per}(B) = e^{h_B} \text{per}(Y^*)$. The symmetric dilation E_b constructed in Section 3.2 has entries of modulus at most $2 \max\{c_{\text{hi}}^2, 1\}/n$ and norm at most $1 - \kappa_b$. Thus Theorem 3.1 applies with $\beta = 2 \max\{c_{\text{hi}}^2, 1\}$ and $\kappa = \kappa_b$. The analysis in Appendix D justifies evaluating the scaling and the truncated formula to polynomial accuracy, after which we restore e^{h_B} . Bounded exceptional dimensions are evaluated exactly. $\square$

B.4. Degree and codegree bounds for bipartite supports.

Proof of Corollary 3.7. Let $Y_{ij}^* = b_{ij} r_i c_j$ be the gauge-fixed Sinkhorn scaling, and write $R_r = r_{\max}/r_{\min}$ and $R_c = c_{\max}/c_{\min}$. Comparing the equations at rows with extremal factors, using their common neighborhood and then interchanging rows and columns, gives

$$R_r \leq \theta^{-1} + \lambda_\theta R_c, \quad R_c \leq \theta^{-1} + \lambda_\theta R_r, \quad \lambda_\theta = \frac{d_+ - \eta}{\theta d_-}.$$

Here the first inequality follows from $1/r_{\min} \leq 1/(\theta r_{\max}) + (d_+ - \eta)mc_{\max}$ and $r_{\max}c_{\min} \leq (\theta d_-m)^{-1}$. Put $R = \max\{R_r, R_c\}$. The assumption (28) says that $\lambda_\theta < 1$. The two preceding inequalities give $R \leq 1/\theta + \lambda_\theta R$, and hence $R \leq 1/(\theta(1 - \lambda_\theta)) = d_-/(\eta - d_+ + \theta d_-)$. Thus

$$(64) \quad \max\{R_r, R_c\} \leq R_\theta, \quad R_\theta := \frac{d_-}{\eta - d_+ + \theta d_-}.$$

The row equations at the minimum and maximum row factors give $1/(d_+m) \leq r_{\min}c_{\max}$ and $r_{\max}c_{\min} \leq 1/(\theta d_-m)$. Using (64), every supported entry of Y^* therefore satisfies

$$(65) \quad \frac{\theta}{R_\theta d_+ m} \leq Y_{ij}^* \leq \frac{R_\theta}{\theta d_- m}.$$

The gauge gives a common geometric mean s_{geom} for the row and column factors. The ratio and extremal-product bounds imply $1/(R_\theta d_+ m) \leq s_{\text{geom}}^2 \leq R_\theta/(\theta d_- m)$ and therefore $1/(R_\theta^{3/2} \sqrt{d_+ m}) \leq r_i, c_j \leq R_\theta^{3/2}/\sqrt{\theta d_- m}$.

Finally, put $W = Y^*(Y^*)^\top$ and $a_\theta := \min\{1/2, \eta\theta^2/(R_\theta^2 d_+^2)\}$. The codegree hypothesis and (65) give $W_{ii'} \geq a_\theta/m$ for distinct rows. The same quadratic-form argument as in the fixed-margin proof yields $\sigma_2(Y^*)^2 \leq 1 - a_\theta$ and hence $\sigma_2(Y^*) \leq 1 - a_\theta/2$. The uniform factor bounds and this singular-value gap verify Theorem 3.4.

For the random-matrix assertion, set $\theta = 1$, choose $\rho = p/8$, and take $\eta = p^2/2$. Then $\eta > d_+ - d_-$. Chernoff bounds give the required row degrees, column degrees, and codegrees with probability $1 - o(1)$. To verify total support, condition on a fixed entry $B_{ij} = 1$ and delete row i and column j . The remaining matrix is an independent Bernoulli(p) matrix of order $s = m - 1$. The Hall estimate (63) gives failure probability $e^{-\Omega_p(m)}$. A union bound over the m^2 possible entries shows that, with probability $1 - o(1)$, every positive entry belongs to a supported permutation. $\square$

APPENDIX C. AUXILIARY ANALYTIC ESTIMATES

This appendix contains the two scalar-factor calculations, the analytic-continuation argument, Gamma-tail localization, cutoff calculus, and matching resolvent estimates deferred from the main text.

C.1. The matching scalar factor.

Proof of Lemma 3.16. Write $z = a + ib$. The inequality $\log u \leq u - 1$, applied to $u = (1 + a)^2 + b^2$, gives $\log |1 + z| \leq a + (a^2 + b^2)/2$. Hence $\log |g(z)| \leq a^2$, with the value at $z = -1$ understood by continuity. When $|z| \leq 1/2$,

$$\log g(z) = \log(1 + z) - z + \frac{z^2}{2} = \sum_{j \geq 3} \frac{(-1)^{j+1}}{j} z^j$$

has modulus at most $2|z|^3$. It follows that $|g(z) - 1| \leq 2e^{1/4}|z|^3 \leq 3|z|^3$. $\square$

C.2. The Ising scalar factor.

Proof of Lemma 4.1. Write $z = x + iy$ with $x, y \in \mathbb{R}$. The identity and envelope estimate are

$$\begin{aligned} |\cosh(x + iy)|^2 &= \sinh^2 x + \cos^2 y \leq \cosh^2 x, \\ |g_{\text{Is}}(x + iy)| &\leq e^{-(x^2 - y^2)/2} \cosh x \leq e^{y^2/2}. \end{aligned}$$

Here the second line uses $\log \cosh x \leq x^2/2$. For the local bound, put $r = |z|$ and $d_k = (2^k k!)^{-1} - (2k)!^{-1} > 0$ for $k \geq 2$. Since $d_2 = 1/12$ and $d_k \leq [12 \cdot 2^{k-2} (k-2)!]^{-1}$ for $k \geq 3$, comparison of the

two power series gives

$$\begin{aligned} |\cosh z - e^{z^2/2}| &\leq \frac{r^4}{12} e^{r^2/2}, \\ |g_{\text{Is}}(z) - 1| &\leq \frac{r^4}{12} e^{r^2} \leq \frac{r^4}{9} \leq \frac{r^4}{8} \quad (r \leq 1/2), \end{aligned}$$

where $e^{1/4} \leq 4/3$. The same series begins with $-z^4/12$, which proves (50) and the stated local constants. $\square$

C.3. Complex Gaussian marginalization.

Proof of Lemma 2.5. The assertion is immediate for $S = \emptyset$, so assume that S is nonempty. In the full representation, write $A_K x = u + iv$, where $u = K_+^{1/2} x$ and $v = K_-^{1/2} x$. Let $\zeta = \sigma + i\tau$. For every $\eta > 0$, the inequality $2|rs| \leq \eta r^2 + \eta^{-1} s^2$ gives

$$\begin{aligned} \sum_{i \in S} (a_{\text{R}}(\text{Re}(\zeta \phi_i))^2 + a_{\text{I}}(\text{Im}(\zeta \phi_i))^2) \\ \leq (1 + \eta) \sigma^2 x^\top K_{\text{env}} x + (1 + \eta^{-1}) \tau^2 x^\top K' x, \end{aligned}$$

where $K' = a_{\text{R}} K_- + a_{\text{I}} K_+ \preceq a_{\text{max}} |K|$. Indeed, $\text{Re}(\zeta \phi_i) = \sigma u_i - \tau v_i$ and $\text{Im}(\zeta \phi_i) = \sigma v_i + \tau u_i$, and extending the sums from S to all coordinates can only increase the two nonnegative quadratic forms.

Choose $\eta, \eta' > 0$ so that $(1 + \eta)(1 + \eta')^2(1 - d) < 1$. Since $\|K\| \leq \|K\|_{\text{row}} \leq \beta$, choose $\omega > 0$ so small and define

$$\begin{aligned} q &:= (1 + \eta)(1 + \eta')^2(1 - d) + 2(1 + \eta^{-1})\omega^2 a_{\text{max}} \beta < 1, \\ \Omega &:= \{\zeta \in \mathbb{C} : |\text{Re } \zeta| < 1 + \eta', |\text{Im } \zeta| < \omega\}. \end{aligned}$$

This is a connected open neighborhood of $[0, 1]$. By (4), $2K_{\text{env}} \preceq (1 - d)I$, while $K' \preceq a_{\text{max}} \beta I$. Uniformly on the closure of every slightly smaller rectangle inside Ω ,

$$|F(\zeta(\phi_i)_{i \in S})| \leq A \exp\left(\frac{q}{2} \|x\|_2^2\right).$$

After multiplication by the standard Gaussian density, the right-hand side is integrable. Thus $\zeta \mapsto \mathbb{E}_K F(\zeta(\phi_i)_{i \in S})$ is holomorphic on Ω .

For the restricted representation put $L = K[S]$. Interlacing gives $\lambda_{\text{max}}(L) \leq \lambda_{\text{max}}(K)$ and $\lambda_{\text{min}}(L) \geq \lambda_{\text{min}}(K)$. The condition $2K_{\text{env}} \preceq (1 - d)I$ bounds $2a_{\text{R}} \lambda_{\text{max}}(K)_+$ and $2a_{\text{I}}(-\lambda_{\text{min}}(K))_+$ by $1 - d$. Hence $2(a_{\text{R}} L_+ + a_{\text{I}} L_-) \preceq (1 - d)I$. Also $\|L\| \leq \|K\| \leq \beta$, so the restricted expectation is holomorphic on the same neighborhood Ω .

The Taylor coefficient of order r at zero of either holomorphic function is a finite linear combination of moments of homogeneous polynomials of degree r . Wick's rule expresses each such moment only through the bilinear second moments, which are $K[S]$ in both representations. The two Taylor series therefore agree. The identity theorem on Ω gives equality at $\zeta = 1$, and the same majorants prove absolute convergence. $\square$

C.4. Localization of the Gamma average.

Proof of Proposition 3.15. Use the notation of Section 3.3. Put $\rho_E = \|E\|$. The entrywise hypothesis gives $\|E\|_{\text{F}} \leq \beta$ and hence $\text{tr } |E| \leq \beta \sqrt{n}$. For every $r > \rho_E$,

$$\mathbb{E}_x \exp\left(\frac{\mathcal{E}_E(x)}{2r}\right) = \det(I - |E|/r)^{-1/2} \leq \exp\left\{\frac{\text{tr } |E|}{2(r - \rho_E)}\right\}.$$

In particular, this is $e^{O(\sqrt{n})}$ at $r = r_0$ and $r = r_1$, uniformly over the matrices in the proposition.

If $U \geq r_1$, then (37) and $1+z \leq e^z$ give $|Z_E(U^{-1})| \leq \mathbb{E}_x e^{\mathcal{E}_E(x)/(2r_1)}$. The Gamma Chernoff bound therefore gives

$$\mathbb{E} [|Z_E(U^{-1})| \mathbf{1}_{\{U \geq r_1\}}] \leq \exp \left\{ -\frac{n}{2}(r_1 - 1 - \log r_1) + O(\sqrt{n}) \right\}.$$

For $0 < u \leq r_0$, the tangent-line inequality for the logarithm gives

$$\left(u + \frac{\mathcal{E}_E(x)}{n} \right)^{n/2} \leq r_0^{n/2} \exp \left\{ \frac{n(u - r_0)}{2r_0} + \frac{\mathcal{E}_E(x)}{2r_0} \right\}.$$

Since $a_n - 1 - n/2 = (\tau - 1)/2$, integrating (37) against (38) bounds the lower-tail contribution by

$$\frac{b_n^{a_n}}{\Gamma(a_n)} r_0^{n/2} \mathbb{E}_x e^{\mathcal{E}_E(x)/(2r_0)} \int_0^{r_0} u^{(\tau-1)/2} \exp \left\{ -b_n u + \frac{n(u - r_0)}{2r_0} \right\} du.$$

Write the exponent in the integral as $-n/2 + \lambda_n u$, where $\lambda_n = n/(2r_0) - b_n = \Omega(n)$. It is increasing for all sufficiently large n . On $[r_0/2, r_0]$, the factor $u^{(\tau-1)/2}$ is bounded, and endpoint integration gives

$$\int_{r_0/2}^{r_0} u^{(\tau-1)/2} e^{-n/2 + \lambda_n u} du \leq \frac{C}{\lambda_n} e^{-n/2 + \lambda_n r_0} = \frac{C}{\lambda_n} e^{-b_n r_0}.$$

On $(0, r_0/2]$, the exponential is at most $e^{-n/2 + \lambda_n r_0/2} = e^{-b_n r_0 - \lambda_n r_0/2}$. The factor $u^{(\tau-1)/2}$ is integrable at zero for both $\tau \in \{0, 1\}$, while $\lambda_n r_0/2 = \Omega(n)$. This part is exponentially smaller than the upper-half contribution. The integral is therefore at most a fixed multiple of $e^{-b_n r_0}$. Stirling's formula now gives

$$\mathbb{E} [|Z_E(U^{-1})| \mathbf{1}_{\{U \leq r_0\}}] \leq \exp \left\{ -\frac{n}{2}(r_0 - 1 - \log r_0) + O(\sqrt{n} + \log n) \right\}.$$

Both rate functions are positive. Decreasing c and increasing n_0 proves (39). $\square$

C.5. The least truncation cutoff.

Proof of Lemma 2.10. Put $D = C_2^{1/\alpha}$ and $A_\alpha = 2 + 2/\alpha$. Choose $\eta > 0$ so small that $A_\alpha \eta \leq c_{\text{tr}}$ and $DA_\alpha \eta \leq 1/4$. The inequality (25) is equivalent to $(DR/n)^R \leq 2^{-b/\alpha}$. Consider

$$R_* = \max \left\{ 2, \left\lceil \frac{2b}{\alpha} \right\rceil \right\}.$$

Since $b \geq 1$ and $b < \eta n$, we have $R_* \leq A_\alpha b < A_\alpha \eta n \leq c_{\text{tr}} n$. Moreover, $DR_*/n \leq 1/4$ and $R_* \geq 2b/\alpha$, so $(DR_*/n)^{R_*} \leq 4^{-R_*} \leq 2^{-b/\alpha}$. Thus a least admissible R exists and is at most R_* .

For the enumeration bound, define $F(x) = x \log(n/(Dx))$. On $2 \leq x \leq R_*$, we have $Dx/n \leq 1/4$, so $F'(x) = \log(n/(Dx)) - 1 \geq \log 4 - 1 > 0$. If $R > 2$, minimality gives $\alpha F(R-1) < b \log 2$. The mean-value theorem and the fixed value of D give $F(R) - F(R-1) = O(\log n)$, and hence $F(R) = O(b + \log n)$. The same conclusion is immediate when $R = 2$. Also $F(R) \geq R \log 4$, so $R = O(b + \log n)$. Finally, $R \log(en/R) = F(R) + R \log(eD) = O(b + \log n)$, which proves (26). $\square$

C.6. Matching resolvent estimates.

Proof of Lemma 3.18. We prove the spectral, entrywise, and determinant bounds in that order. Eigenvalue calculus handles the first, the resolvent expansion the second, and the identity $\text{tr } E = 0$ the third.

If e is an eigenvalue of E , the corresponding eigenvalue of K_t is $te/(1+te)$. This function is increasing, and $|te| \leq q_0$. We may therefore take $d = (1 - q_0)/(1 + q_0)$.

Put $\rho = \|E\|$, and let $\mathbf{e}_i$ denote the i th coordinate vector. For $\ell \geq 2$, the entrywise hypothesis on E gives

$$|(E^\ell)_{ij}| \leq \|E\mathbf{e}_i\|_2 \|E\|^{\ell-2} \|E\mathbf{e}_j\|_2 \leq \frac{\beta^2}{n} \rho^{\ell-2}.$$

Expanding K_t as a convergent resolvent series proves the entrywise bound in (41). For example, one may take $\beta' = \beta/r_0 + (\beta^2/r_0^2)/(1 - q_0)$.

Finally, $\text{tr } E = 0$ because E has zero diagonal. If $\lambda_1, \dots, \lambda_n$ are its eigenvalues, then $\log \det(I + tE) = \sum_i (\log(1 + t\lambda_i) - t\lambda_i)$. The summands have absolute value at most $t^2\lambda_i^2/(2(1 - q_0))$. Since $\|E\|_F \leq \beta$, their sum is bounded by a constant. $\square$

APPENDIX D. BIT COMPLEXITY

The main text describes the algorithms in real arithmetic. This appendix records the stability and precision estimates that give the stated bit complexities.

D.1. Perfect matchings. Retain the notation b, R, C, c_*, K_t , and $Z_{E,R}$ from Section 3.5. For a real symmetric matrix F and a scalar u , write $K_{u,F} = uF(I + uF)^{-1}$ and abbreviate $Z_R(t, E) = Z_{E,R}(t)$.

Lemma D.1 (Stability of the truncated formula). *Fix $q_{\text{stab}} < 1$ and $C_F < \infty$. There is a constant C_{stab} with the following property. Let $2 \leq R \leq c_*$, and suppose that every pair (u, F) on the line segment joining (t, E) and $(\tilde{t}, \tilde{E})$ satisfies*

$$\text{tr } F = 0, \quad \|F\|_F \leq C_F, \quad \|uF\| \leq q_{\text{stab}}, \quad \|K_{u,F}[T]\|_{\text{row}} \leq \frac{1}{4} \quad (|T| < R).$$

Then

$$|Z_R(\tilde{t}, \tilde{E}) - Z_R(t, E)| \leq n^{C_{\text{stab}}} \exp\left(C_{\text{stab}} R \log \frac{en}{R}\right) (|\tilde{t} - t| + \|\tilde{E} - E\|_{\max}).$$

The constant C_{stab} depends only on the fixed margins.

Proof. The differential of the transformed matrix is $dK = (I + tE)^{-1}(dt E + t dE)(I + tE)^{-1}$. The resolvent margin and $\|dE\| \leq n\|dE\|_{\max}$ contribute only a polynomial factor. The trace, Frobenius, and spectral assumptions likewise control the determinant factor and its derivative. For a fixed set T , the row-norm condition gives $\|(I - K_{u,F}[T])^{-1}\|_{\text{row}} \leq 4/3$. Differentiating (42) and its monomer-dimer recurrence therefore bounds its value and derivative by $n^{O(1)}e^{O(|T|)}$. Inclusion-exclusion over $T \subseteq S$ contributes a factor $2^{|S|}$. Summing over $|S| < R$ and applying the mean value theorem proves the claim. $\square$

Remark D.2 (Numerical precision for the matching algorithms). Assume that the input and ε are rational, and let L_{in} denote their total encoding length. Writing $r_v = n^{-1/2}e^{z_v}$, the symmetric scaling is the minimizer of

$$\Psi_A(z) = \frac{1}{n} \sum_{uv \in E(G)} a_{uv} e^{z_u + z_v} - \sum_v z_v.$$

The bipartite scaling is obtained similarly by minimizing

$$\Psi_B(x, y) = \frac{1}{m} \sum_{b_{ij} > 0} b_{ij} e^{x_i + y_j} - \sum_i x_i - \sum_j y_j$$

on the gauge subspace $\sum_i x_i = \sum_j y_j$. The factor bounds and spectral gaps in Section 3.2 make these potentials uniformly strongly convex and smooth on fixed boxes containing their minimizers. Standard projected gradient descent therefore finds the scaling to accuracy $2^{-\nu}$ in $O(\nu + \log n)$ iterations [Nes18, Chapter 2].

The entropy exponent is recovered from the same vector with error at most $\sqrt{n}2^{-\nu}$, since $h_A(G) = \frac{n}{2} \log n - \sum_v z_v$ by the proof of Proposition 3.9, and $h_B = m \log m - \sum_i x_i - \sum_j y_j$ likewise. Let h denote the relevant entropy and let $\tilde{h}$ denote its computed approximation. Choose ν so that $|\tilde{h} - h| \leq 2^{-b}$, which changes the final entropy factor by at most $e^{\pm 2^{-b}}$.

Use a fixed $\bar{q}_0 \in (q_0, 1)$ as q_{stab} in the stability lemma. The exact bounds $\|tE\| \leq q_0$ and $\|K_t[T]\|_{\text{row}} \leq 1/8$, together with continuity, imply that every sufficiently accurate approximation and the line segment joining it to the exact pair satisfy $\|uF\| \leq \bar{q}_0$ and $\|K_{u,F}[T]\|_{\text{row}} \leq 1/4$.

By Lemma D.1 and (46), the retained formula has size $n^{O(1)}2^{O(b)}$ and is Lipschitz with a constant of the same form. Its inverses remain uniformly conditioned because $I + tE \succeq (1 - q_0)I$ and $I - K_t[T] \succeq 3I/4$. A gate-by-gate absolute-error induction bounds the accumulated rounding radius by the circuit size times $\exp(O(n \log n))$ times the unit roundoff. No division is made by a signed subset contribution. Thus outward-rounded dyadic evaluation with $O(n \log(n+2) + b + L_{\text{in}})$ working bits makes the scaling, input-perturbation, and rounding errors each at most 2^{-b} . Together with the three analytic errors, these give an additive error in Q_E of at most $6 \cdot 2^{-b} \leq 6\varepsilon/C^2$, hence a relative error of at most $\varepsilon/2$. The entropy error was budgeted separately above.

In the exact branch, the subset recurrence is applied directly to the original matrix A , or to the symmetric dilation of B , rather than to the generally irrational scaling. Clearing denominators once gives intermediate integers of polynomial bit length, and the recurrence returns zero when the hafnian is zero. This proves the bit-complexity bounds in Theorems 1.1, 3.3 and 3.4.

D.2. The Ising model. Retain b, R , and K from the proof of Theorem 1.2. The number of elementary exponential terms is

$$(66) \quad M_R = \sum_{s < R} \binom{n}{s} 3^s, \quad \log M_R = O_{\beta, \kappa}(b + \log n),$$

by (61). The row-norm clause of Theorem 2.3 gives $\|K[T]\|_{\text{row}} \leq 1/8$ for every retained set. Hence Lemma 4.4 applies, its exponential arguments have magnitude $O(R)$, and $I + K[T] \succeq 7I/8$. The rational matrices K , $K[T](I + K[T])^{-1}$, and the relevant determinants have bit length $\text{poly}(n, L_J)$.

Evaluate each elementary term to absolute error at most $2^{-b-2}/M_R$ and the final prefactor to relative error 2^{-b-2} . This requires $\text{poly}(n, L_J) + O_{\beta, \kappa}(b + \log n)$ working bits. Standard algorithms for rational linear algebra, square roots, and exponentials then take polynomial time per term. Together with (66), this proves the bit-complexity bound in Theorem 1.2.